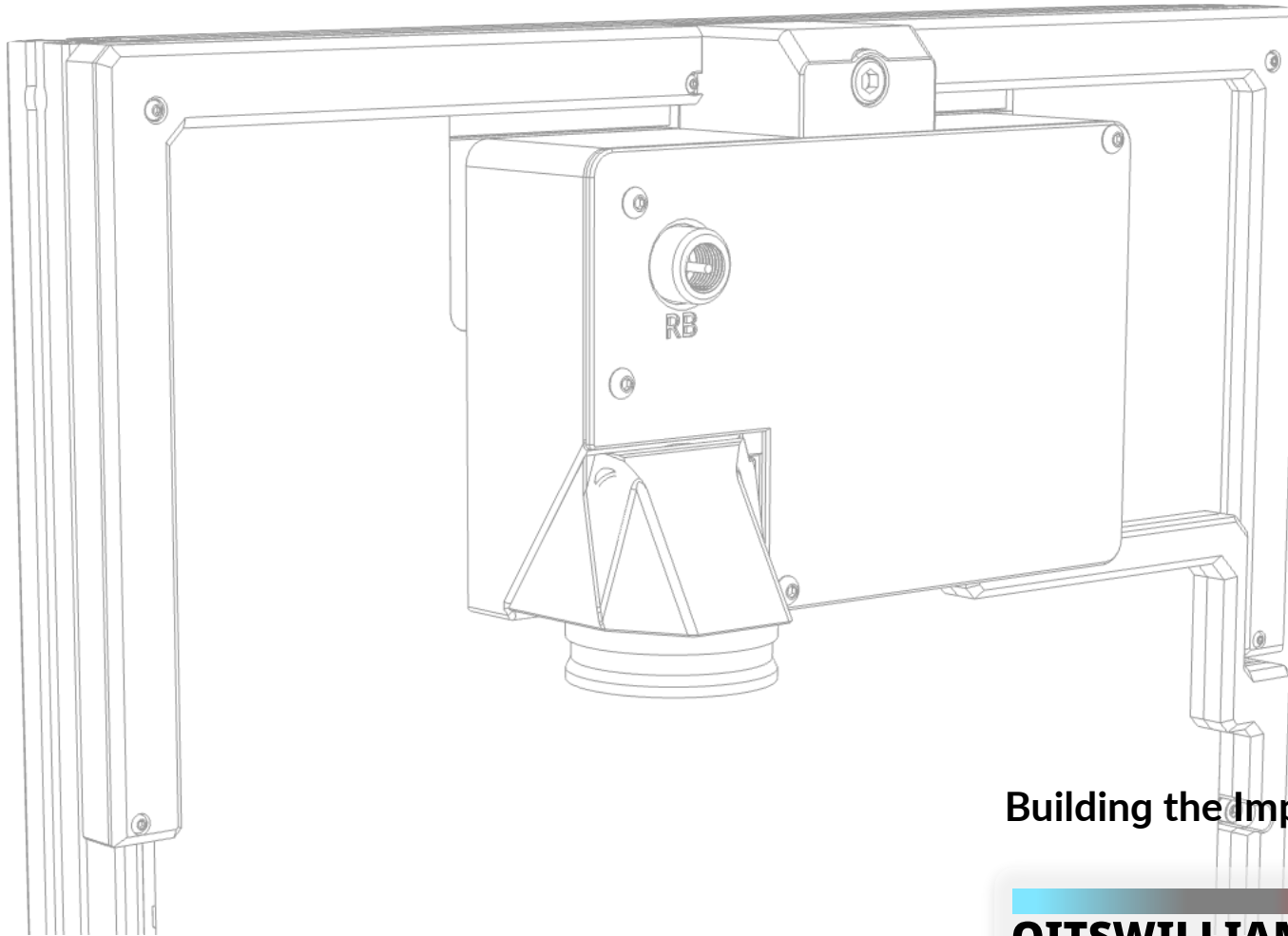


Exhaust Lite
Generation 2
Assembly manual
REV 2024-12-04



Building the Impossible.

OITSWILLIAM PANG

Hardware

Contents

• <u>Before getting started</u>	4
• <u>About (this hardware)</u>	5
• <u>Warnings</u>	6
• <u>Contribution</u>	7
• <u>Support</u>	
• <u>Part printing</u>	8
• <u>Assembly</u>	9
• <u>Inner cover</u>	10
• <u>LED Back</u>	11
• <u>Unit body</u>	18
• <u>Filament runout sensing (BSPP model only)</u>	20
• <u>Inserting the Bowden tube coupler</u>	23
• <u>Attaching the body unit</u>	29
• <u>Attaching inner Bowden tube</u>	31

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• <u>Webcam installation</u>	34
• <u>OV5640 laptop webcam</u>	
• <u>Mini webcam</u>	39
• <u>Cable cover</u>	44
• <u>Webcam wiring</u>	47
• <u>Filter cartridge</u>	48
• <u>Slotting the filter</u>	52
• <u>Attaching the back Bowden tube</u>	54
• <u>Vent port</u>	55
• <u>Software configuration</u>	57

Before getting started

You are about to read the mod project assembly manual.

This manual shows the assembly, wiring and configuration guide for Exhaust Lite Gen 2 for Voron Trident and Voron 2 printers.

Guide and legend

Material

- highlighted material / part

Material

- mentioned material / part

Model
model-
variant-1.stl /
model-
variant-2.stl /
model.step

- printable / moldable model
(these includes STL, 3MF or
sometimes even source files
like STEP)

The model name may not be mentioned
in later pages.

Electronic parts will show their
own power throughout this
assembly guide, but here are the
abbreviations that will be used:

V - volt,

A or I - amp (I as amp may be used
but not in this manual.),

GND - ground.

In DC,

+ - positive,

- - negative

In AC,

L - live

N - neutral

$\perp$, E or G - earth ground

Other abbreviations include:

MCU - microcontroller unit

SBC - single-board computer

About Exhaust Lite

Exhaust Lite is a shrunk-down version of exhaust unit that was made for portability. It uses activated charcoal pellets to purify VOCs made by melting filament. It can last for up to 3 months before it needs to be replaced or reactivated.

Exhaust Lite Gen 2 uses 4010 axial fan and supports external venting hose with 38mm of diameter. This makes print ventilation easier.

Exhaust Lite has array of LEDs called LED back to show the printer stats. It also allows mounting webcam inside the printer.

Warnings



Danger: risk of electric shock

- Even though you can build with this mod from start, if you want to wire, or re-wire, you need to make sure and check that the mains is unplugged.
- Failure doing so which is unplugging the mains power may cause electrocution or even death, especially on regions that use 240V.
- When in doubt, please don't touch the wires connected to the power supply or SSR.

Warning: electrostatic discharge

- You should be wearing with an anti-ESD strap that is connected to earth ground before disassembling certain components.
- You may stop and not wear such strap when accessing high-voltage components.

Contribution

This project is a part of OITSWILLIAMV2 mod pack. If you want to contribute this project, you can visit GitHub.com/Bunny350/OITSWILLIAMV2.

In other words, each Oitswilliam projects have their own licensing policies. So if you want to distribute the projects, you need to refer to their licensing policies first.

Support

Having problems building our projects? You can contact us in Facebook or Twitter.
<https://www.facebook.com/officialbunny350>

However, for the better experience, you can visit and then join Oitswilliam Pang Discord server and receive the community support. Please follow it's rules. <https://discord.gg/Cu6e9ra>

If you have problems because of the project's errors or simply issues, you can submit them in GitHub.
<https://GitHub.com/Bunny350/OITSWILLIAMV2/issues>

The manual might be updated overtime, so when you start building, or are trying to build the projects, you might need to get the latest manual if it was left for at least every three months.

Thanks for viewing the manual. Now go on the next page to build the first part.

Part printing

Print guidelines

Each part of the model requires the following set settings, failure to match most of them may result in unexpected damages.

Bold text indicates the recommended value and *italic text* indicates what setting it is being changed.

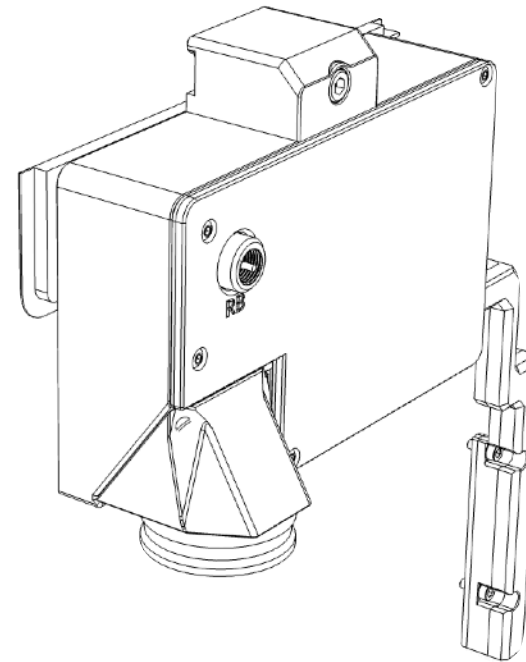
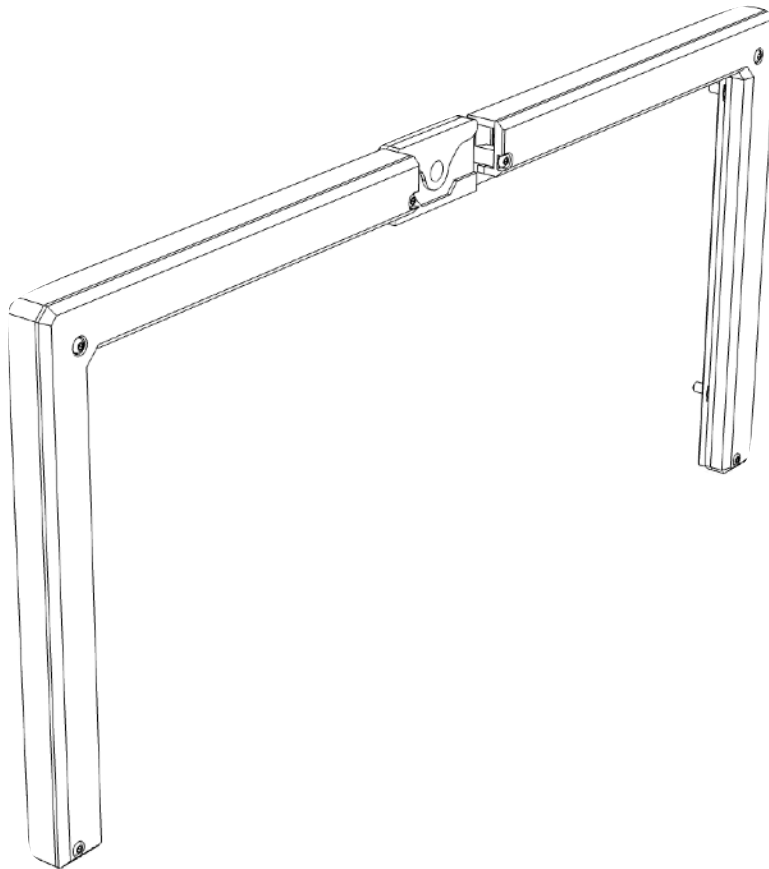
Printer types FDM / FFF or SLS SLA only for parts with no heat-seat parts	Print material: ABS, ASA or PA12 Prohibits PLA.
Layer height of 0.2mm Initial layer height can be 0.2mm or higher	<i>Nozzle and extrusion width</i> must be forced 0.4mm ; if using 0.4mm nozzle
1.6mm wall width Or 4 wall counts , excluding tools and jigs	<i>Infill</i> must be at least 40% , excluding tools and jigs
At least 4 solid bottom and top layers.	The best <i>print speed</i> is up to 120mm/s . The speed might be reduced by the hot end flow rate.
<i>Supports</i> are not required . Uncheck <i>generate support</i> if available.	<i>Other settings might be depended on your printer.</i> <i>Follow the print material warnings for proper handling.</i>

Before assembly

To get started building the project, you may use the following tools:

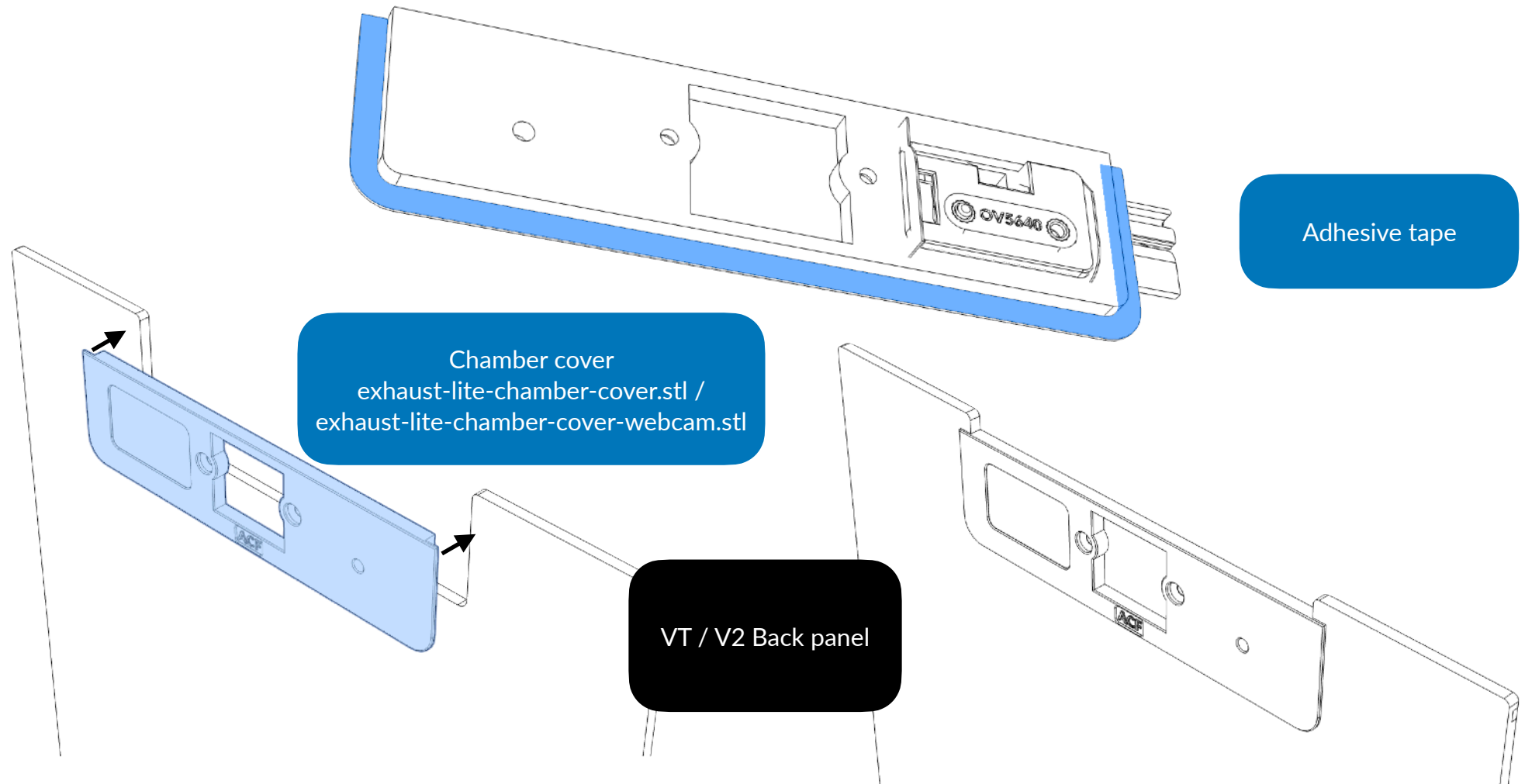
- Soldering iron
- Heat shrink tubes (including red, black, green and yellow, or labeled), at least 1mm diameter
- Superglue (on attaching vent port and magnet)
- Marker and utility knife or specifically made splicer / cutter (for splicing the Bowden tube)

Assembly



Inner cover

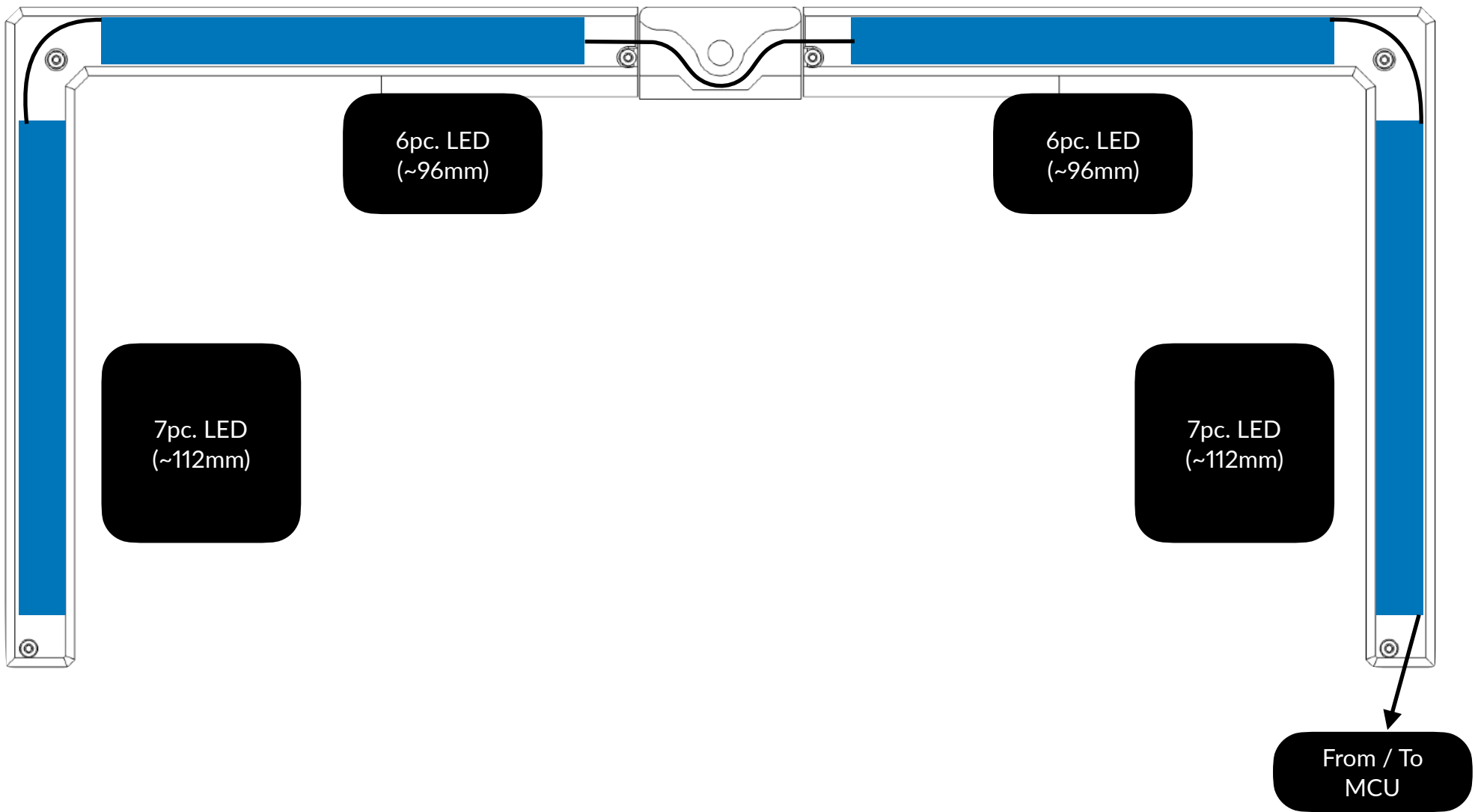
For covers without built-in webcam or webcam attached outside such cover, use exhaust-lite-chamber-cover.stl, otherwise if you want to install webcam into such cover, use exhaust-lite-chamber-cover-webcam.stl.



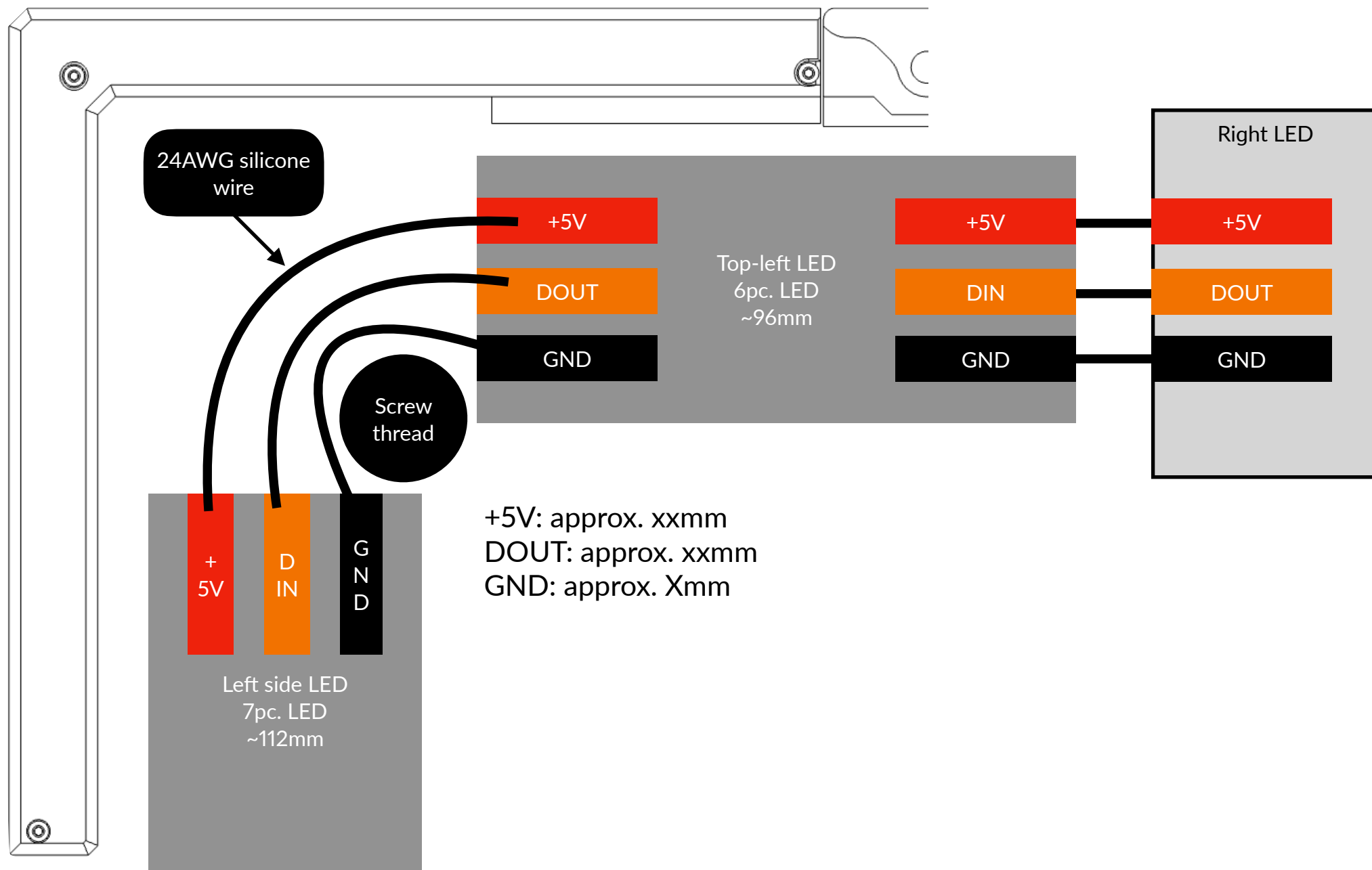
LED back

Wiring reference

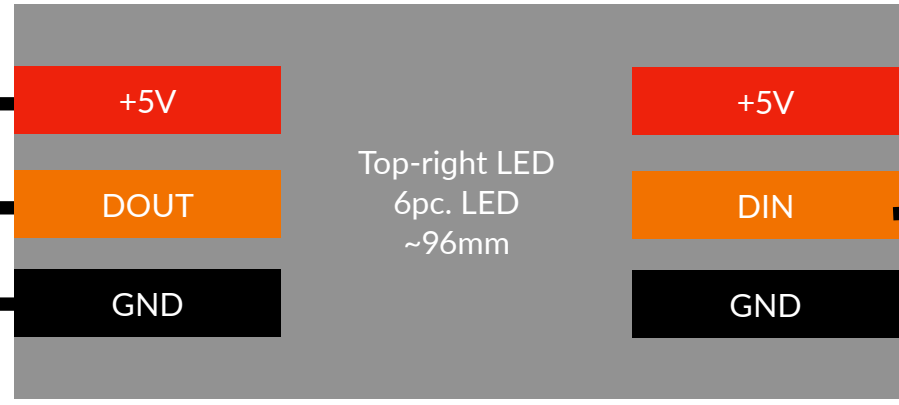
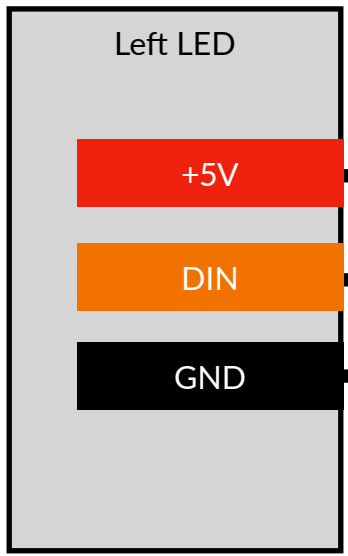
This page shows for 60 LEDs per meter. If this differs like for example the strip has less or more LEDs per meter, you may use approximate dimensions and you may need it longer or shorter, appropriately. In this manual, we will guide it's wiring, then wire before attaching, which is then done by attaching the printed part first.



Left LED



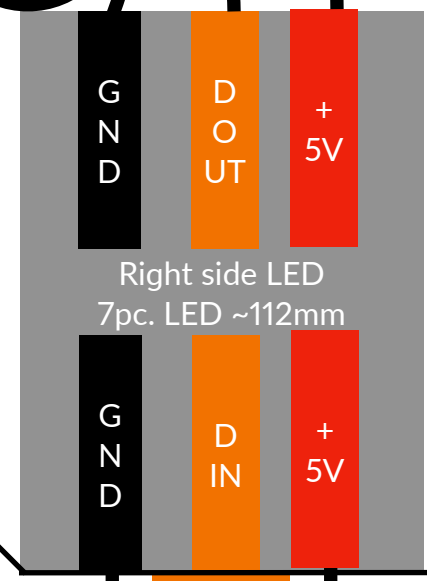
Right LED



24AWG silicone wire

Screw thread

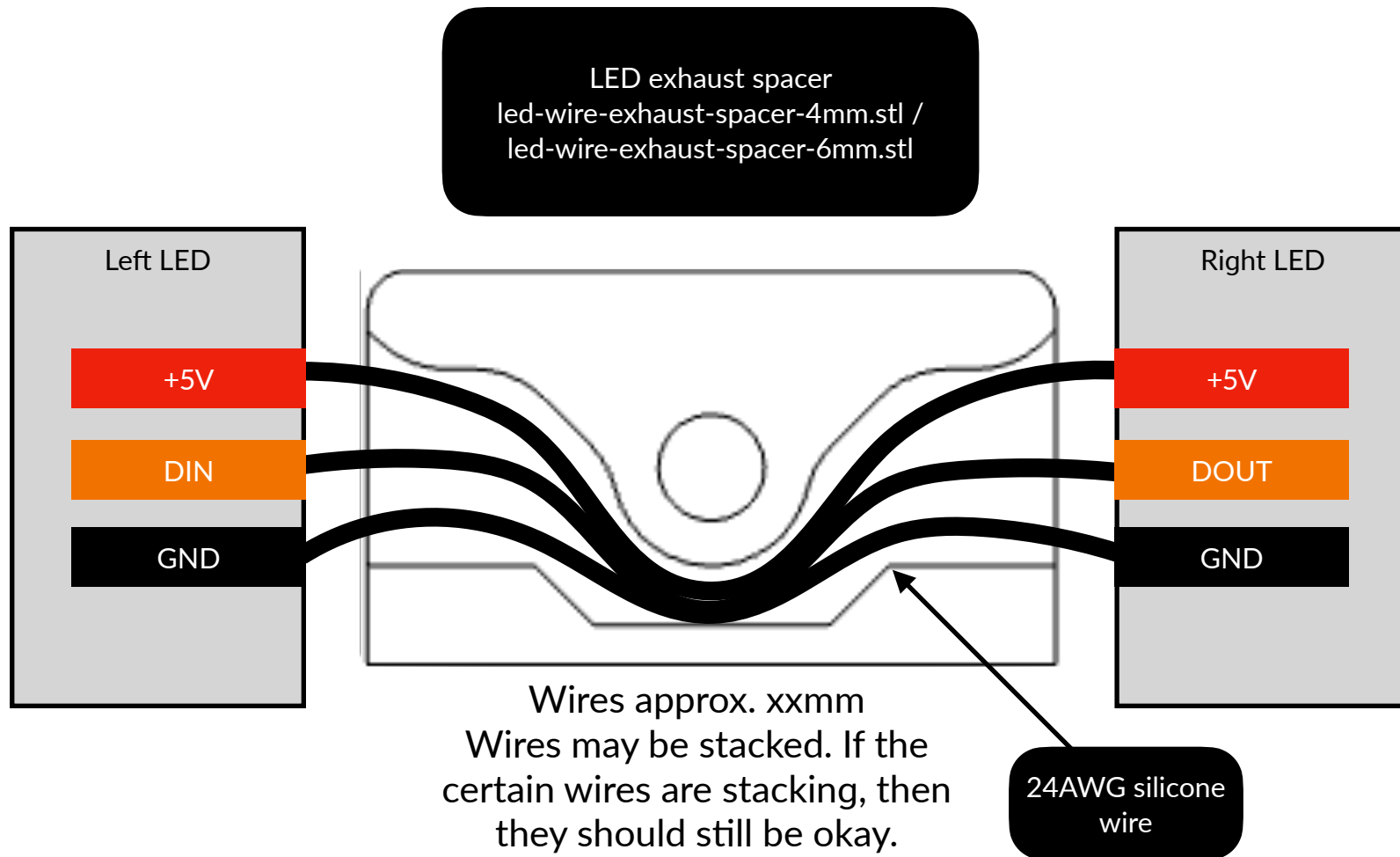
+5V: approx. xxmm
DOUT: approx. xxmm
GND: approx. Xmm



Ending with approx.
Wire (print vol.) /
350mm (150mm)
450mm (250mm)
500mm (300mm)
550mm (350mm)



Center



Mount installation

T-nuts must be preloaded first.

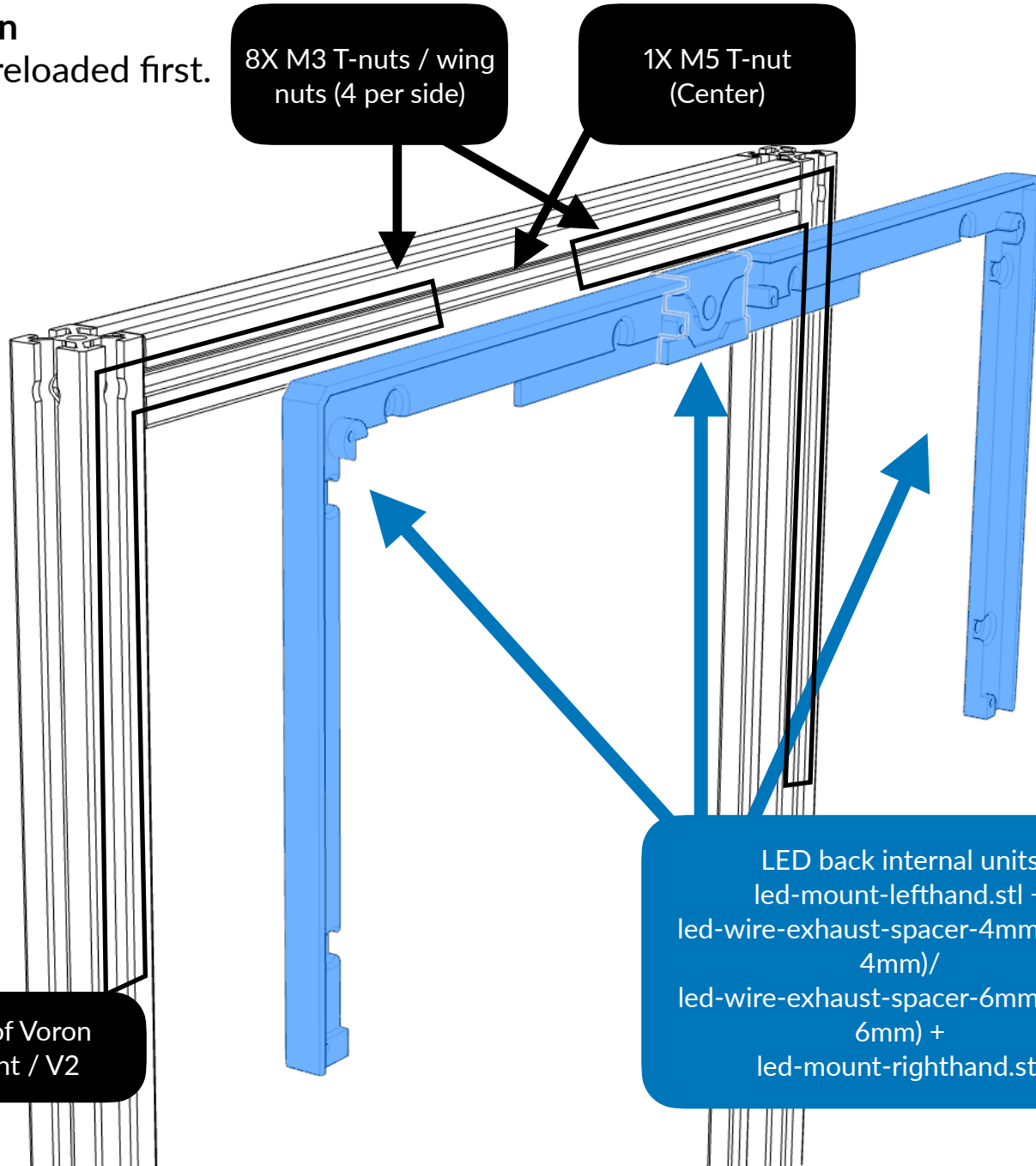
8X M3 T-nuts / wing
nuts (4 per side)

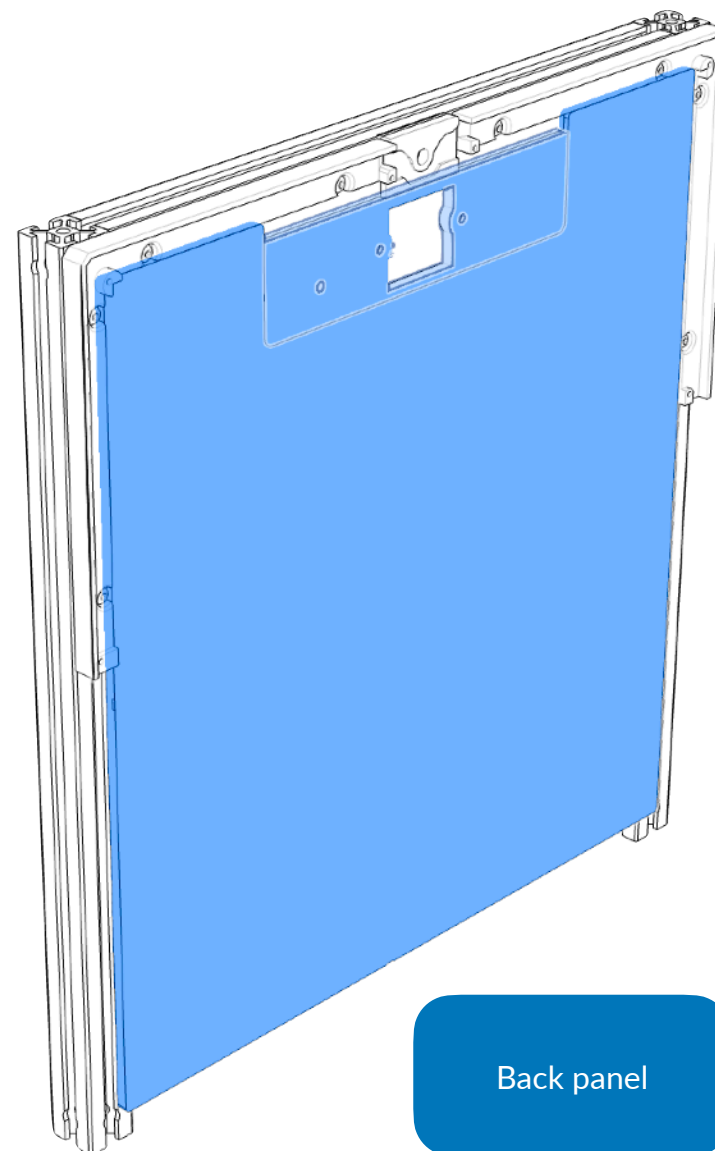
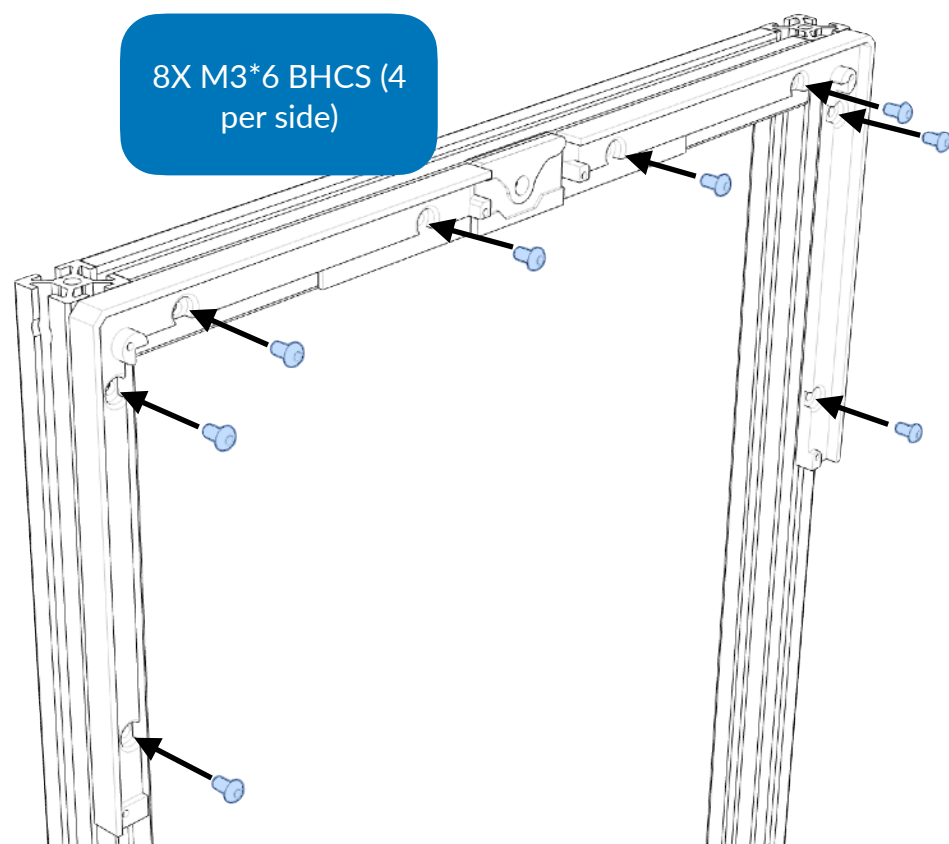
1X M5 T-nut
(Center)

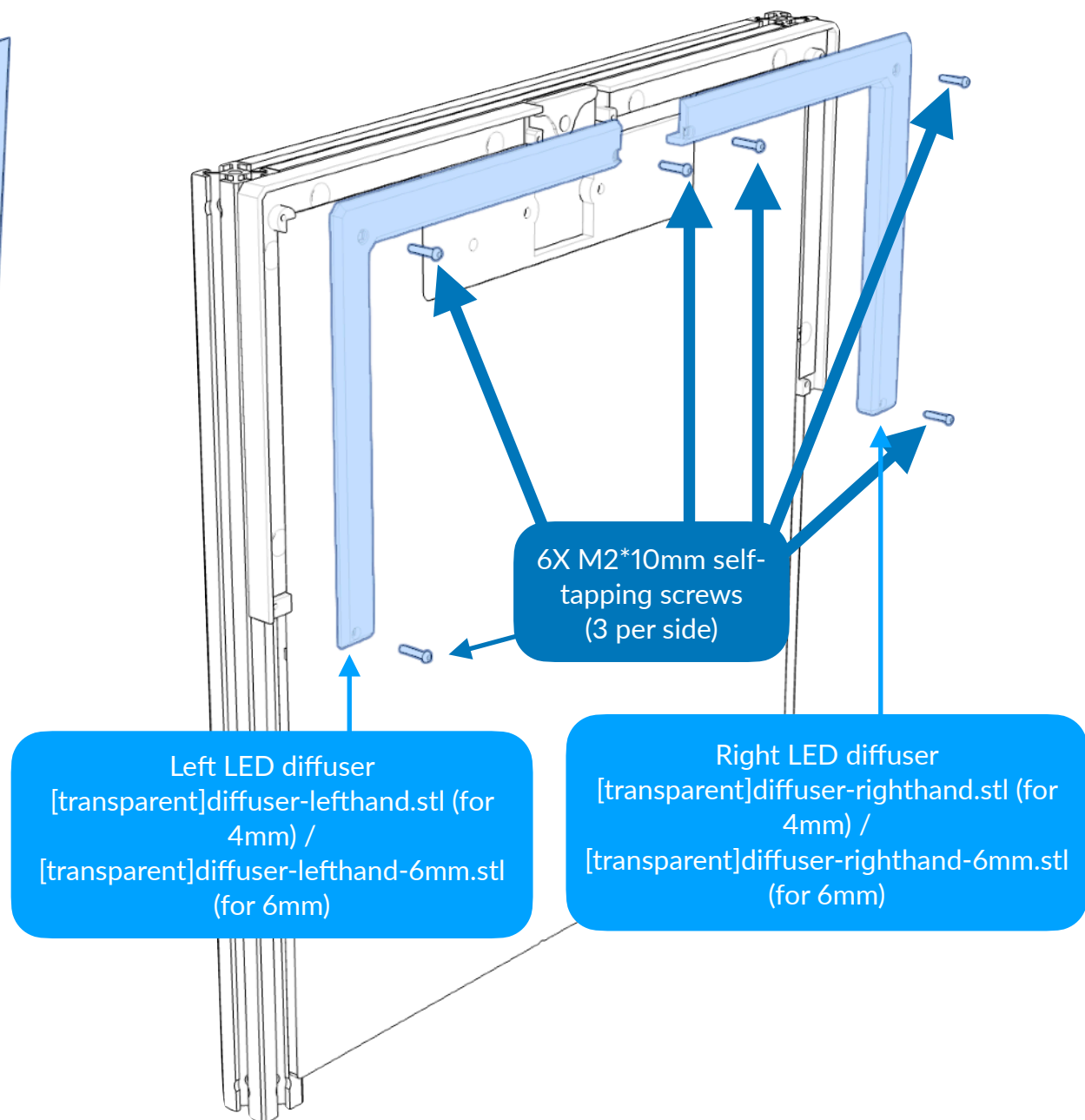
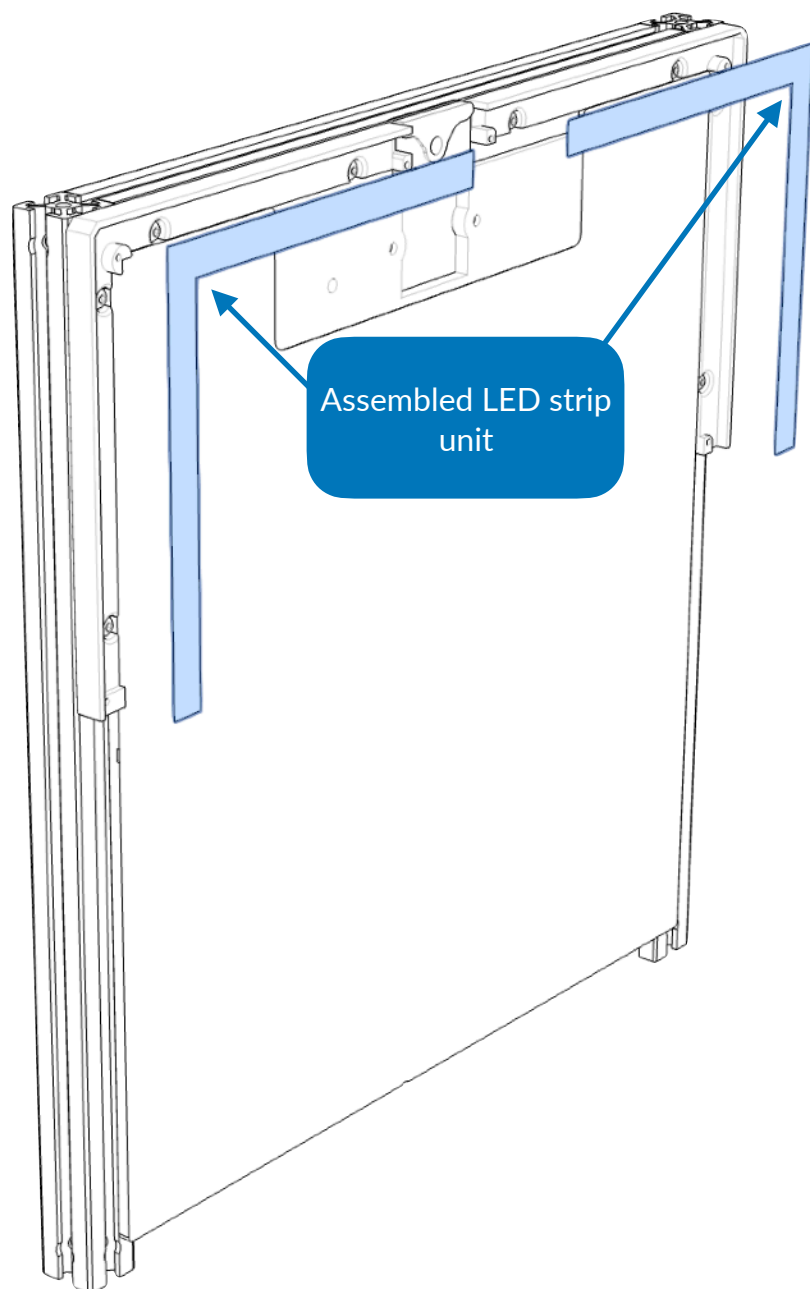
You may rely on the total thickness of the back panel. If the panel has total thickness of 4mm then use `led-wire-exhaust-spacer-4mm.stl`, or if the panel has total thickness of 6mm, then use `led-wire-exhaust-spacer-6mm.stl`. This procedure will continue on the diffusers.

Back of Voron
Trident / V2

LED back internal units
`led-mount-lefthand.stl` +
`led-wire-exhaust-spacer-4mm.stl` (for
4mm)/
`led-wire-exhaust-spacer-6mm.stl` (for
6mm) +
`led-mount-righthand.stl`





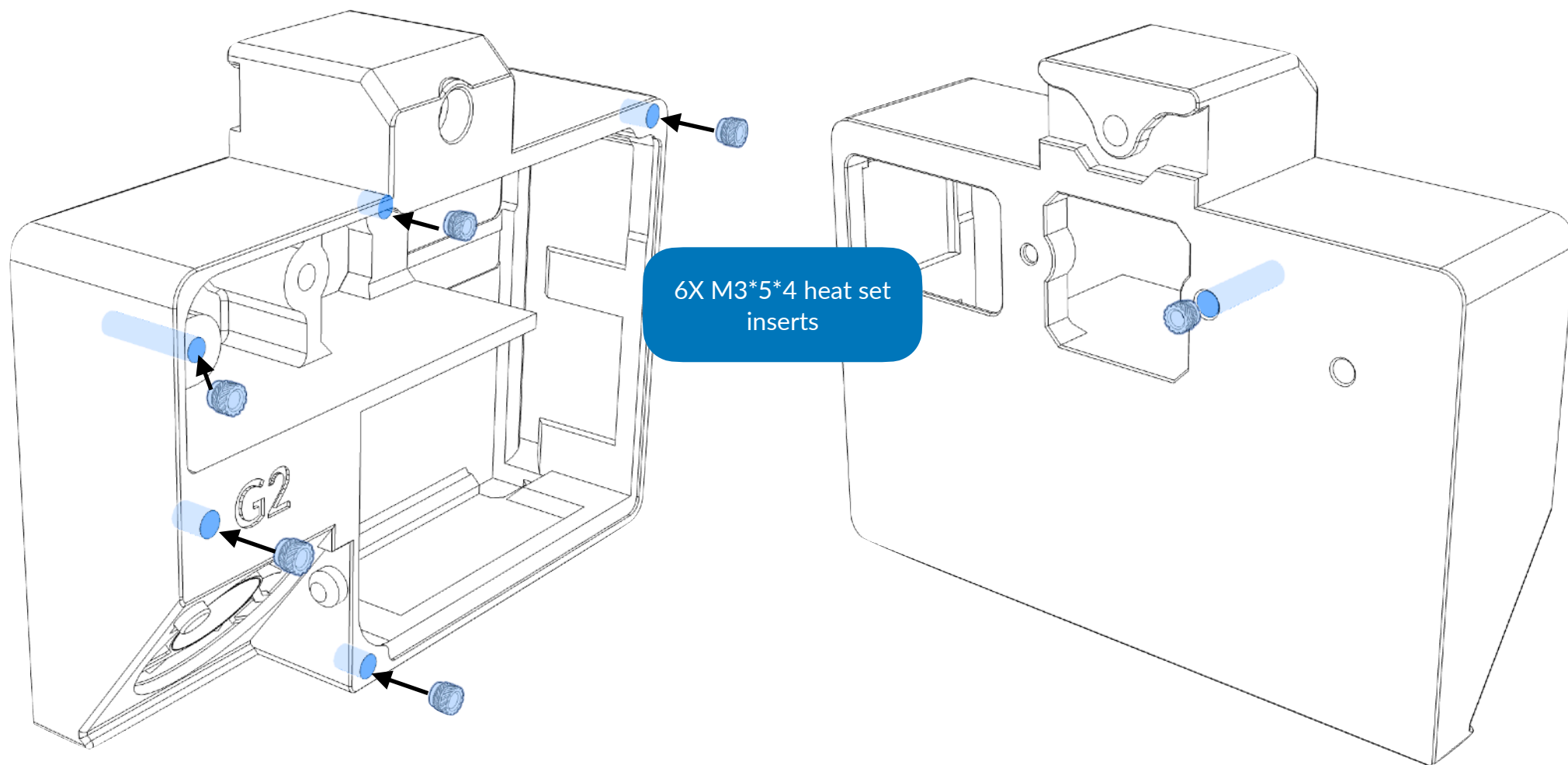


Unit body

Preparation

Main unit body
main-body.stl

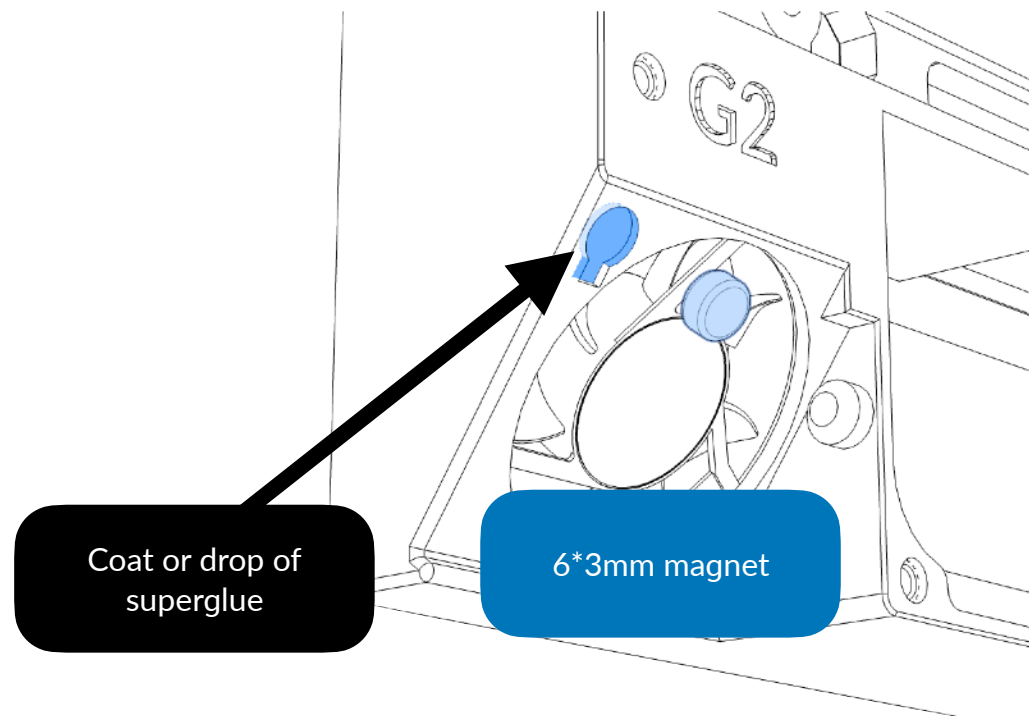
6X M3*5*4 heat set
inserts



Hose attachment magnet

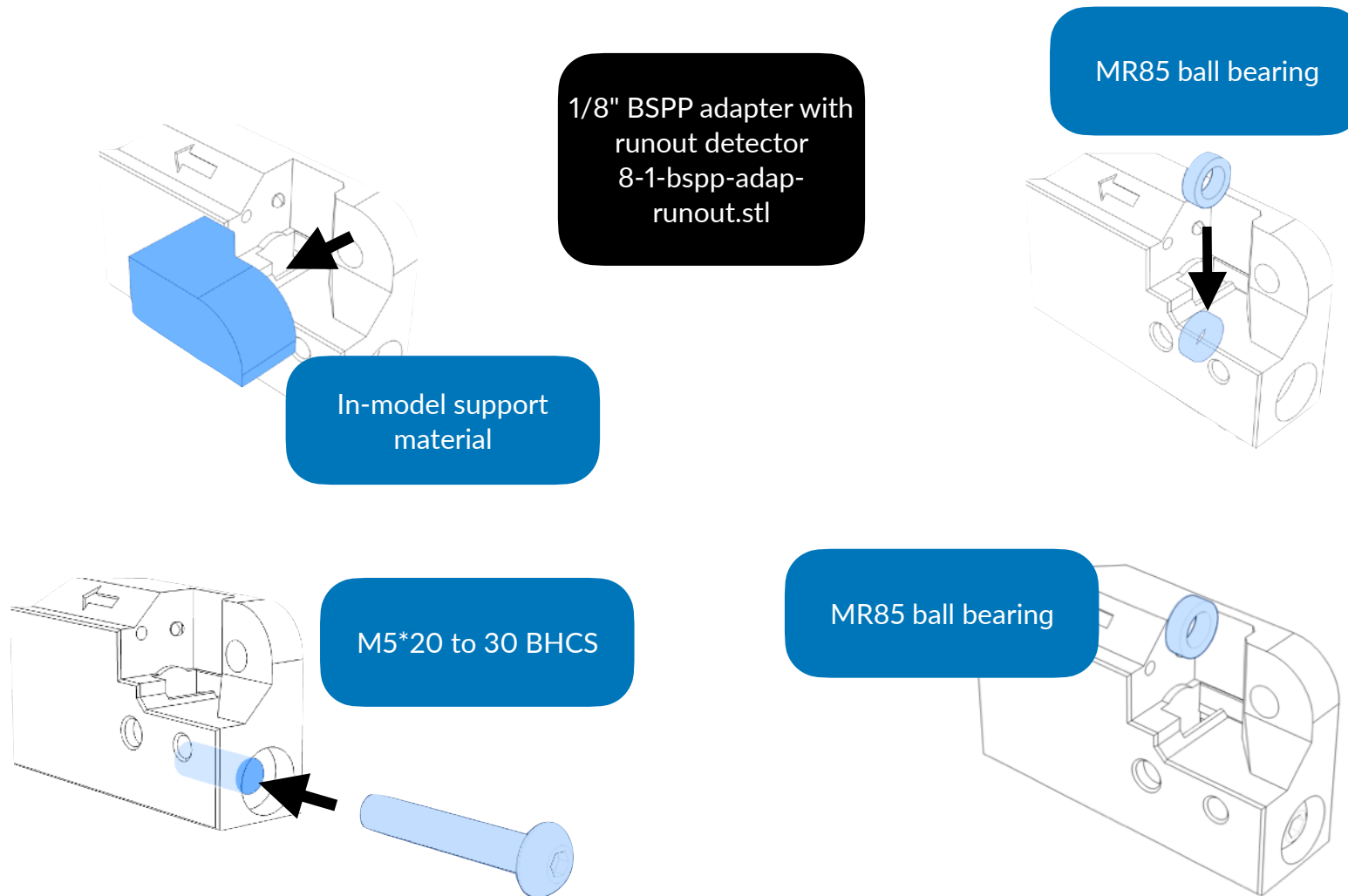
Absolute directions (north or south) doesn't matter as long as they attract each other for single-unit use, otherwise, if it or the hose is being used across somewhere around the room, building or the world, you may stick the magnet with it facing south to the unit side. You may use a U-shape magnet or a bar magnet with color-code and / or north and south letters.

The attachment cutout must be coated with or with at least a drop of superglue.

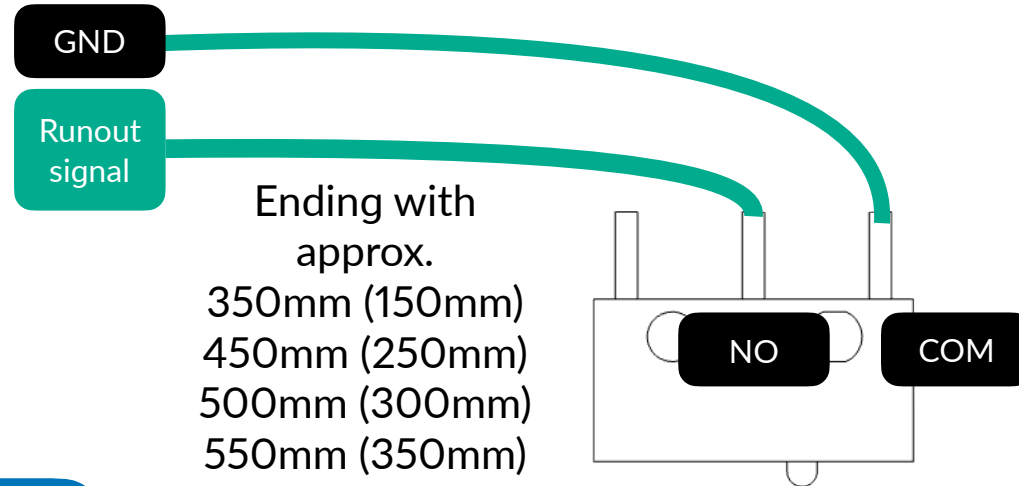


Installing filament runout detector

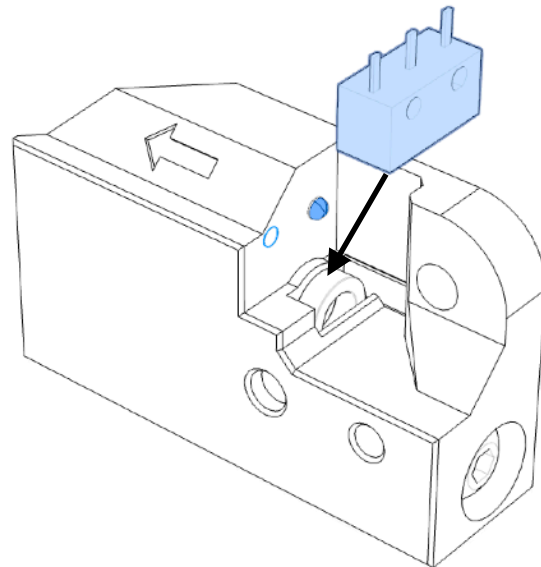
1/8" BSPP only. Only needed if runout sensing from ELG2 is required. May not be used if other runout sensing nearer between the hot-end is attached.



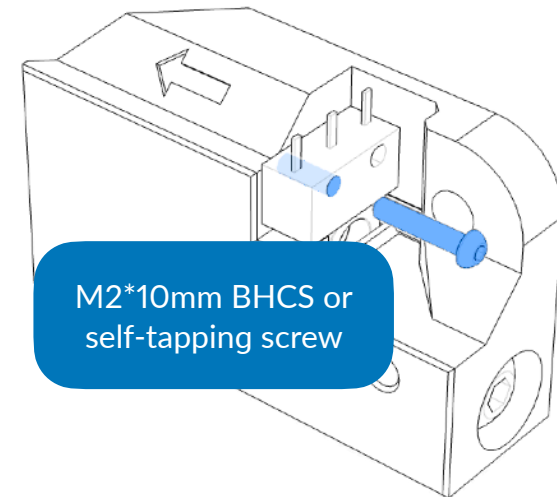
Microswitch wiring and installation



D2F Microswitch

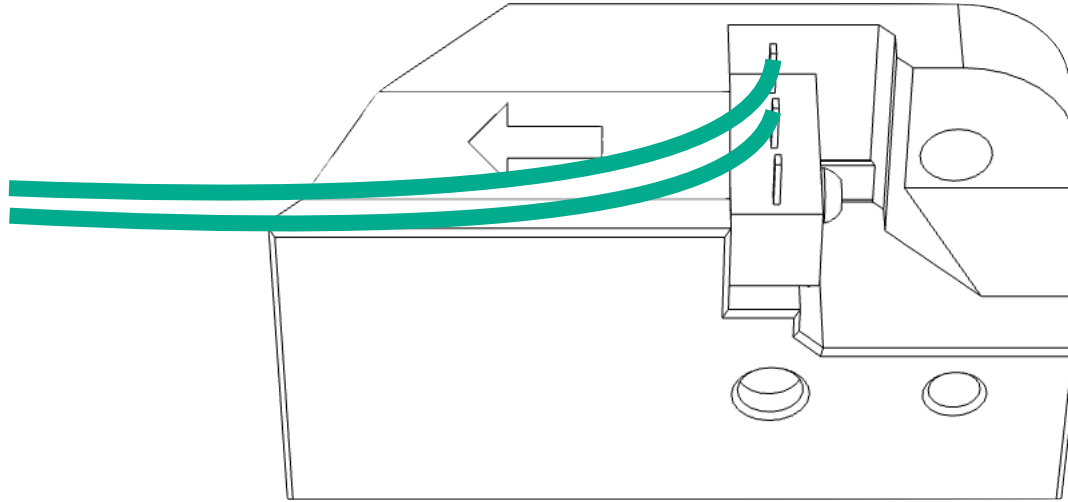


The microswitch lever must be removed before proceeding this step. To remove the lever, you can press the thin side of the lever with pliers and then pull it.



M2*10mm BHCS or self-tapping screw

Tidying up wires

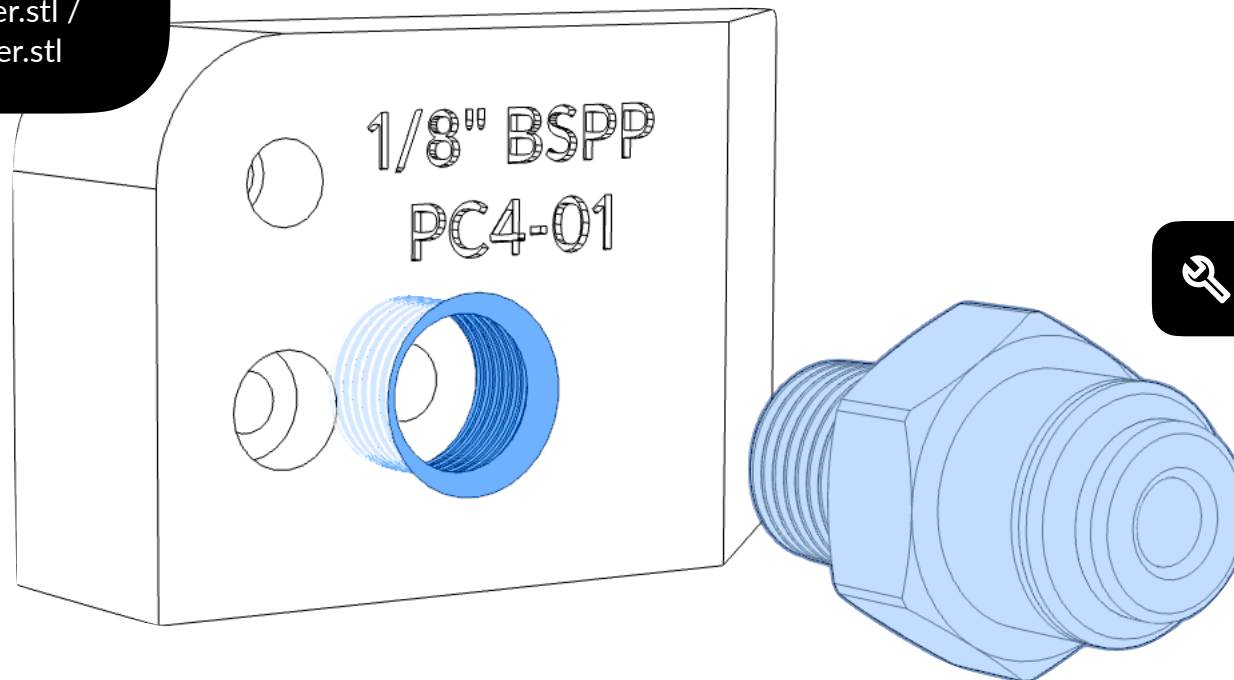


The runout sensor wires must be tucked and placed on the lefthand cutout where the left arrow is pointing at.

Inserting the Bowden tube coupler

You may use the wrench if hand-tightening is too weak.

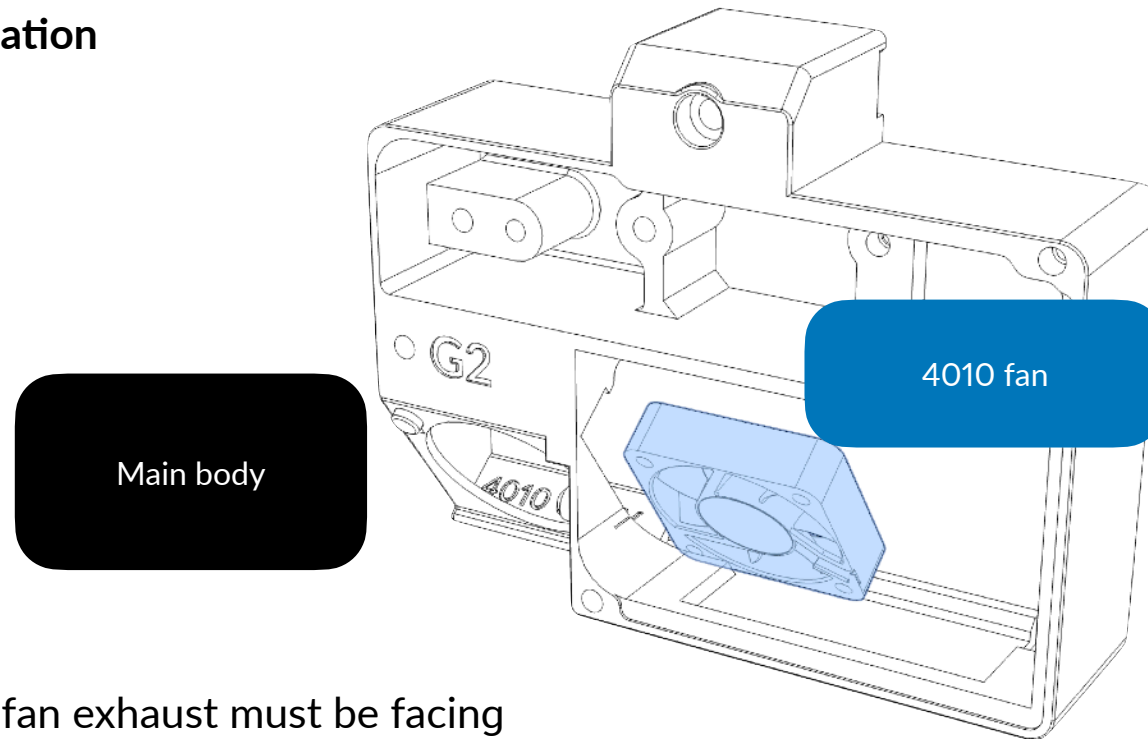
Bowden tube adapter
1/8" BSPP adapter with runout
detector or
8-1-bspp-adapter.stl /
pc4-m6-adapter.stl



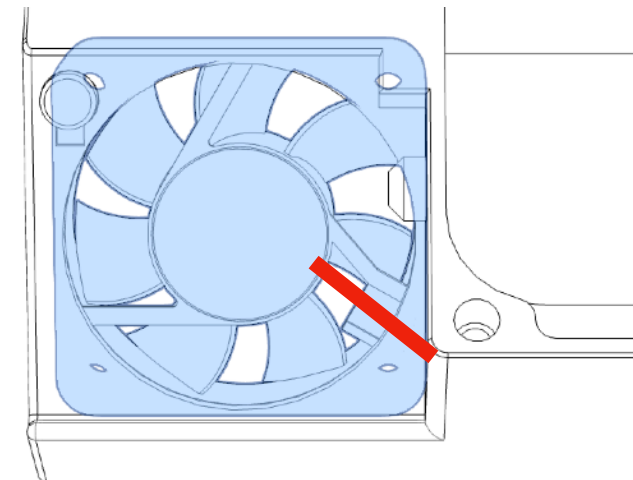
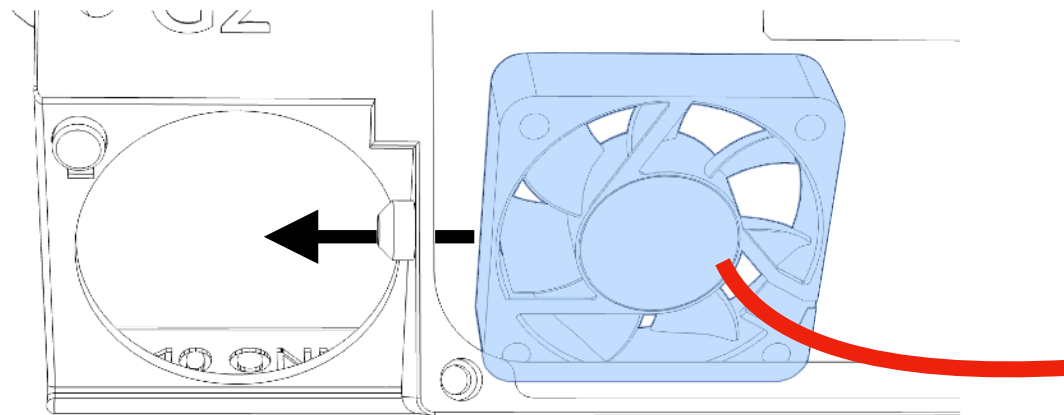
10 or 15mm

Bowden tube coupler

Exhaust fan installation

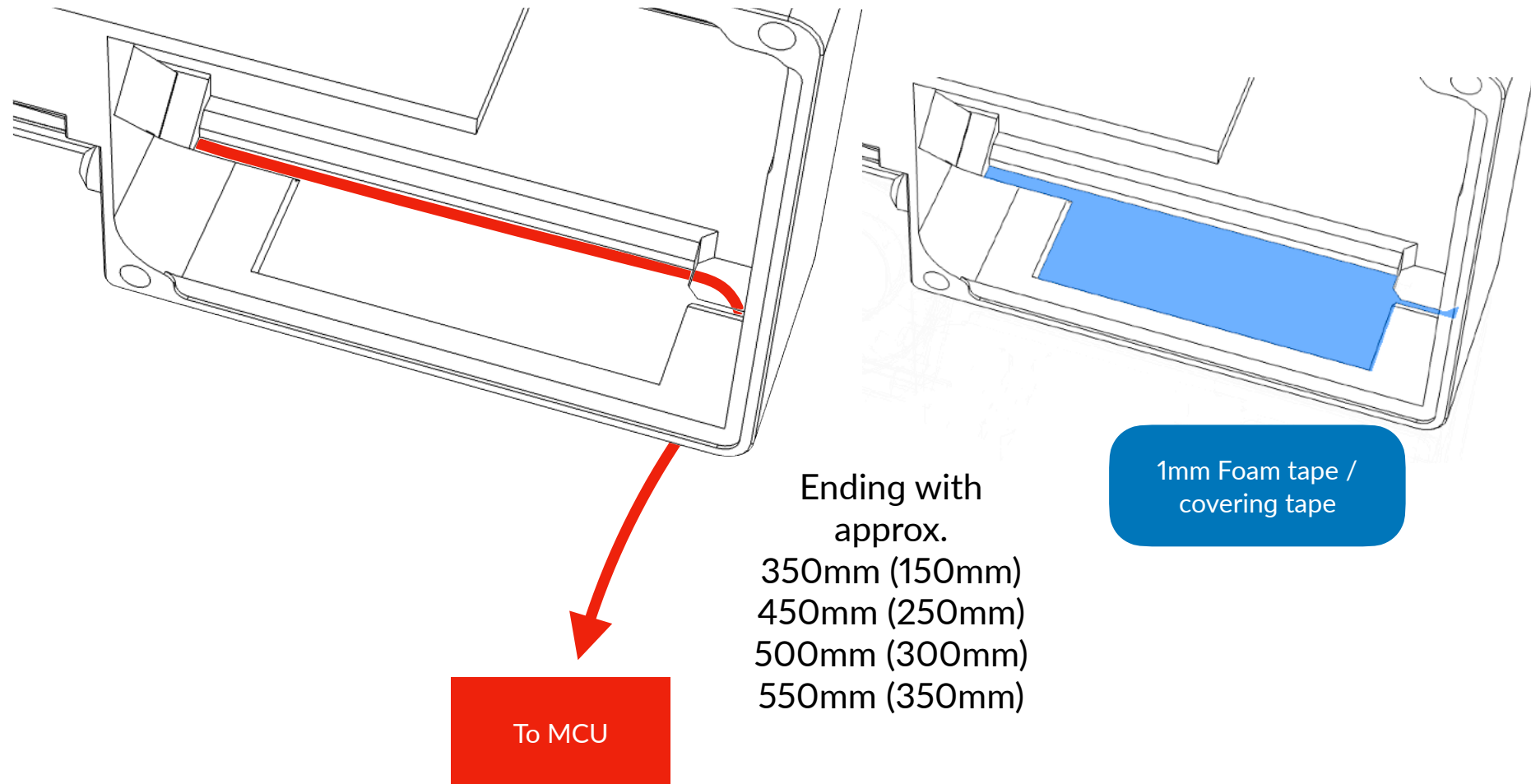


The fan exhaust must be facing back / exposed and the fan wires must be facing bottom.



Wire feeding

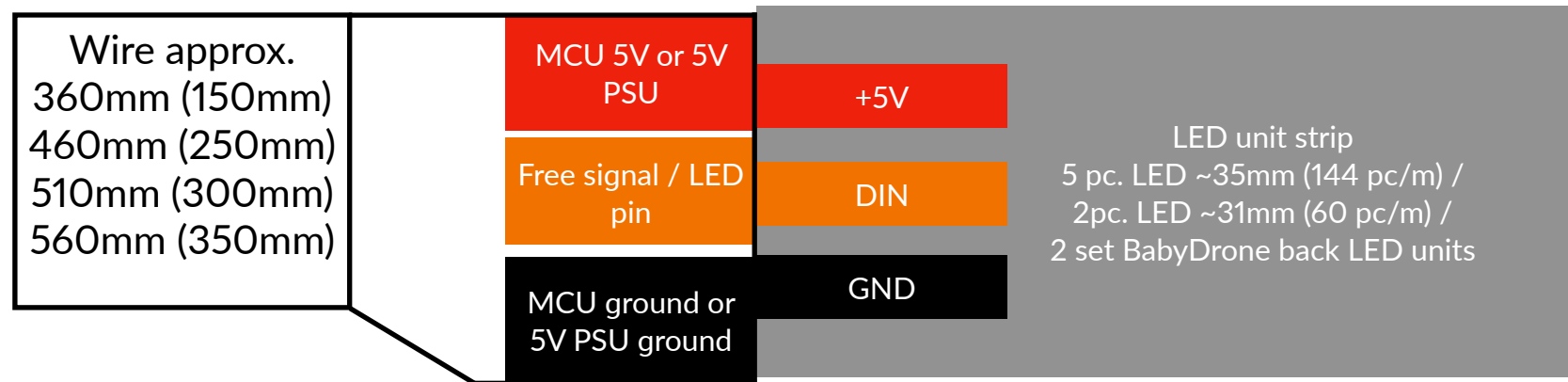
The wire from the fan to MCU must be tucked and then covered by 1mm foam tape or covering tape.



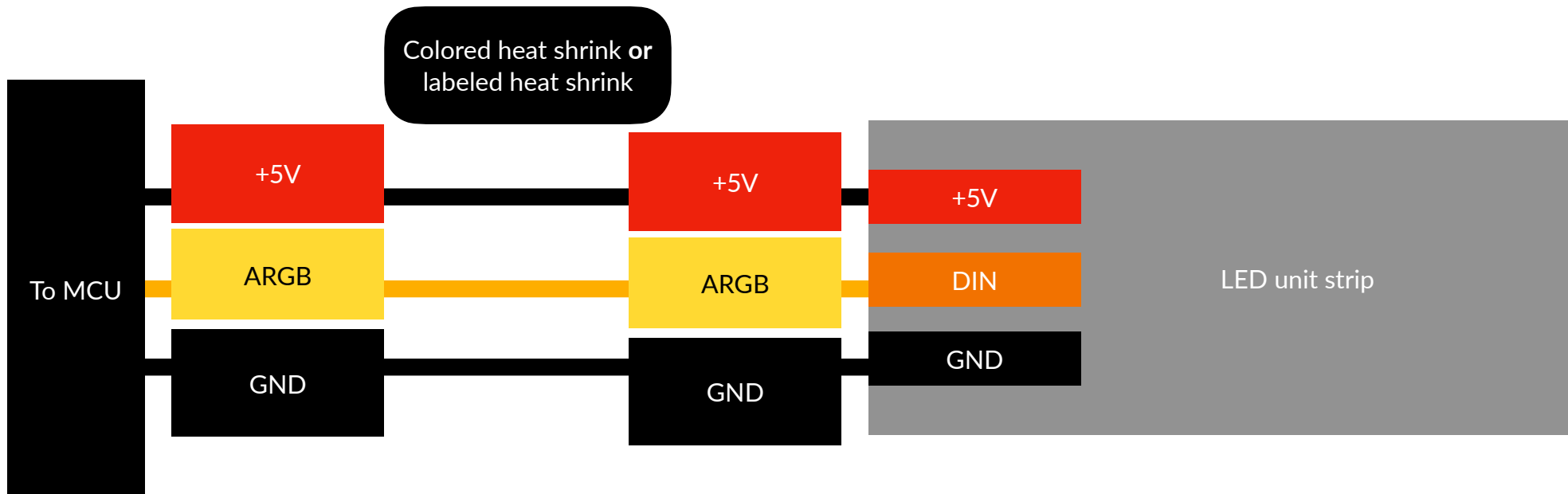
LED unit

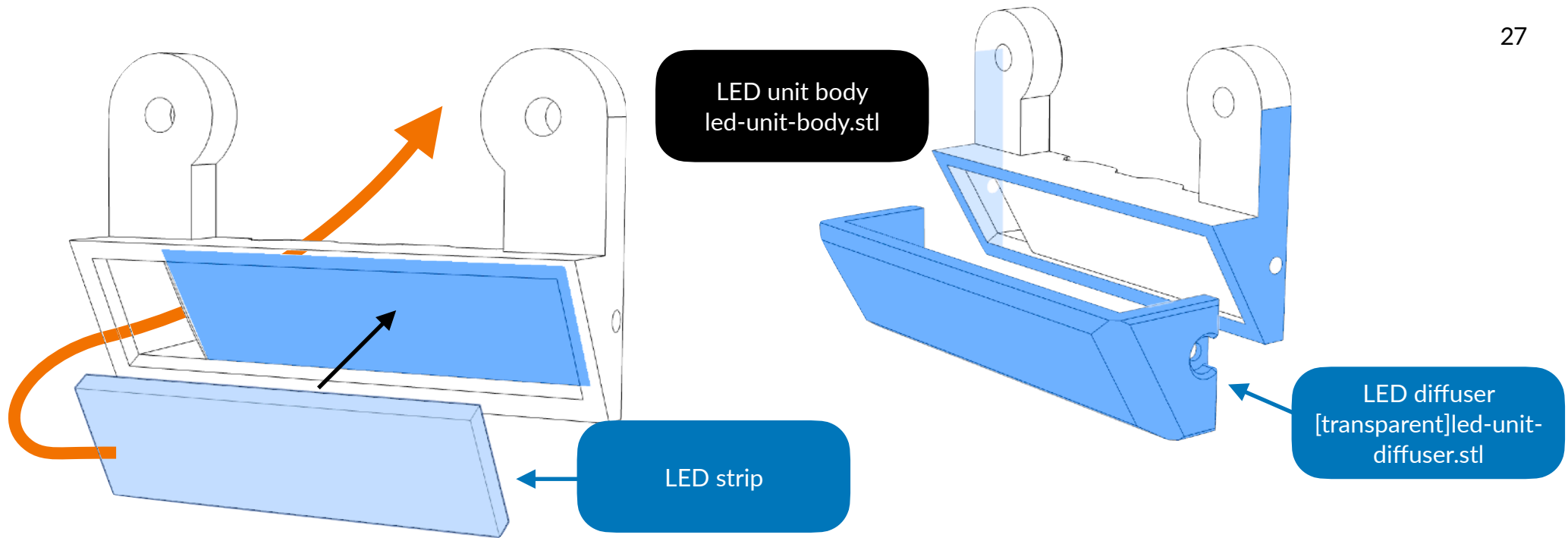
Optional step, might be used if the environment is too dark most of the time.

This unit uses a strip that has 144 LEDs per meter as standard, but you can use the strip with 60 LEDs per meter. Otherwise you may use two OIWP BabyDrone back LED units.

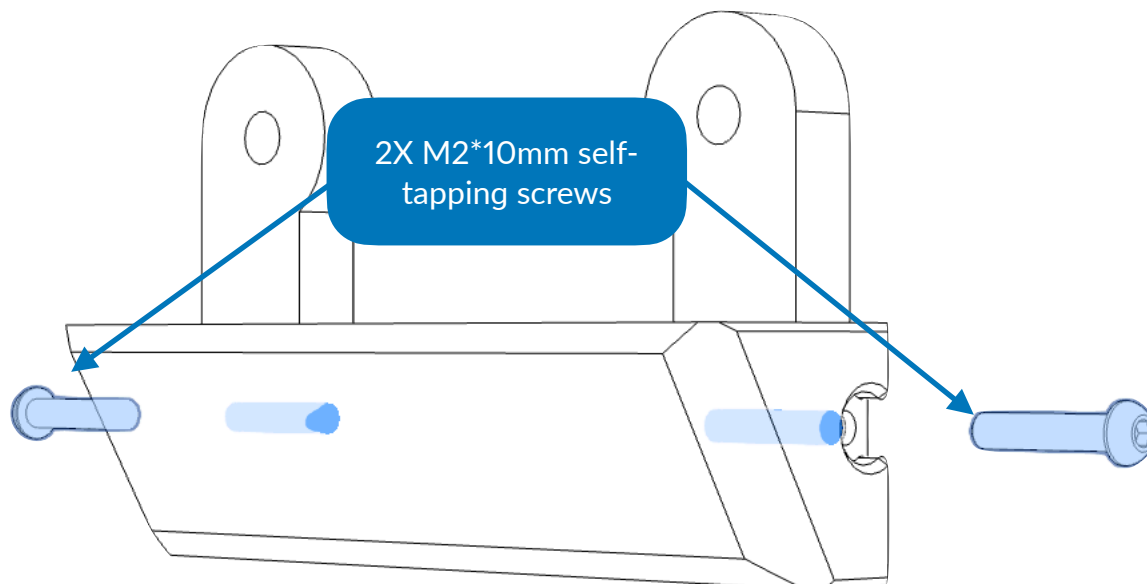


Heat shrink guiding





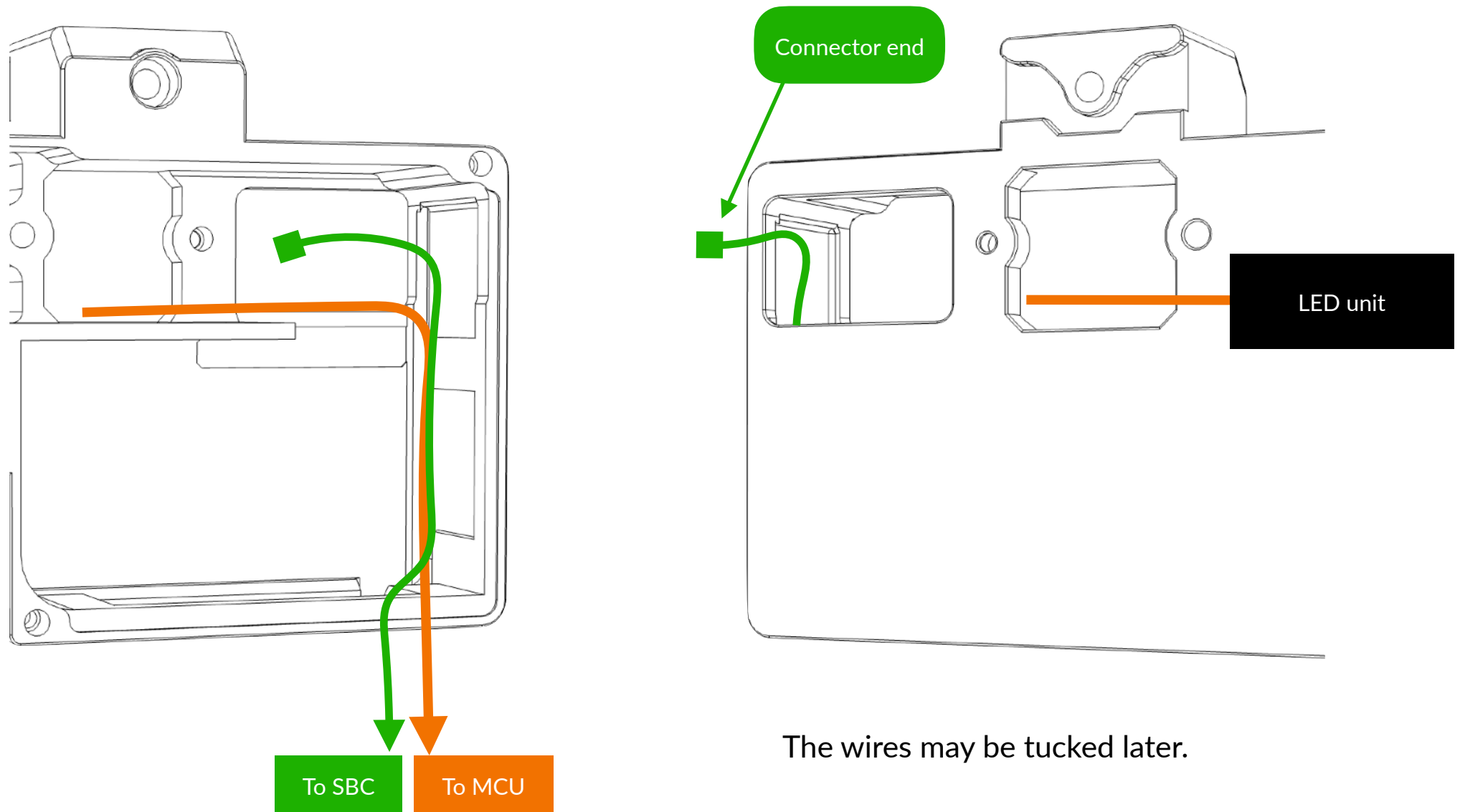
Attaching the diffuser to the unit body



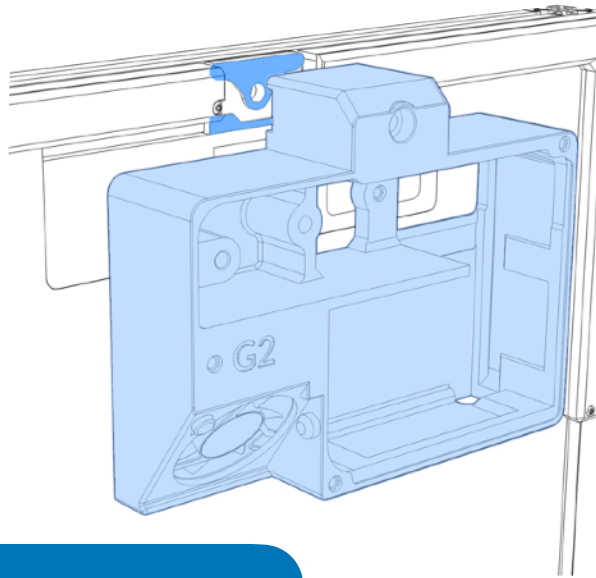
Do not crimp the connectors just yet! You may head to the next page without crimping these connectors.

Webcam cable and chamber LED wires preparation

Reference

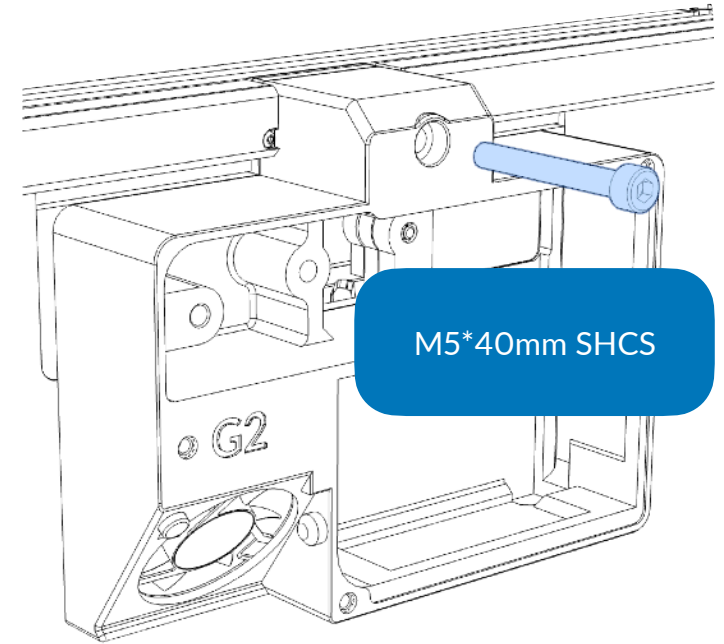


Attaching the body unit

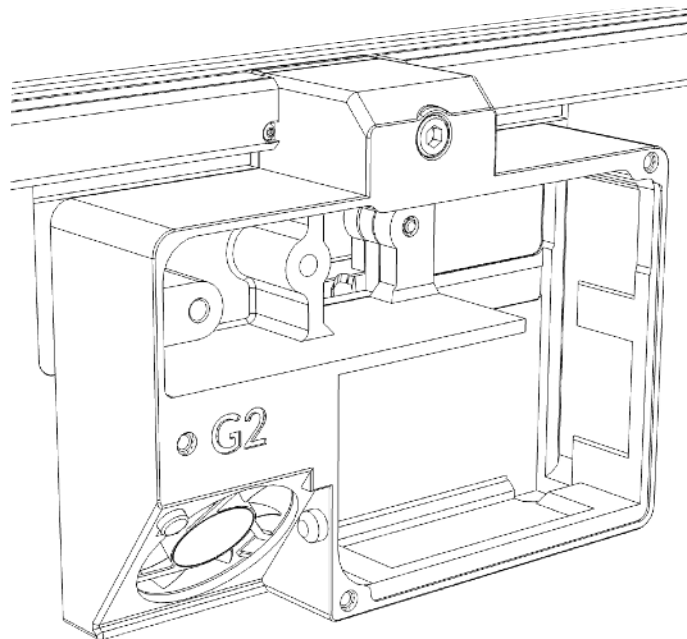


Partially assembled
ELG2 body unit

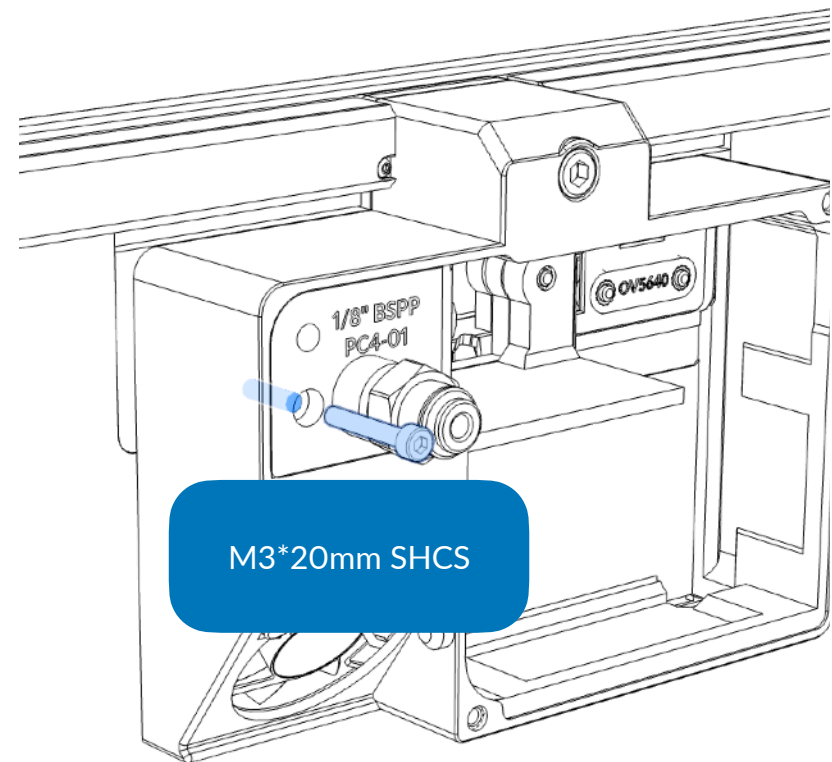
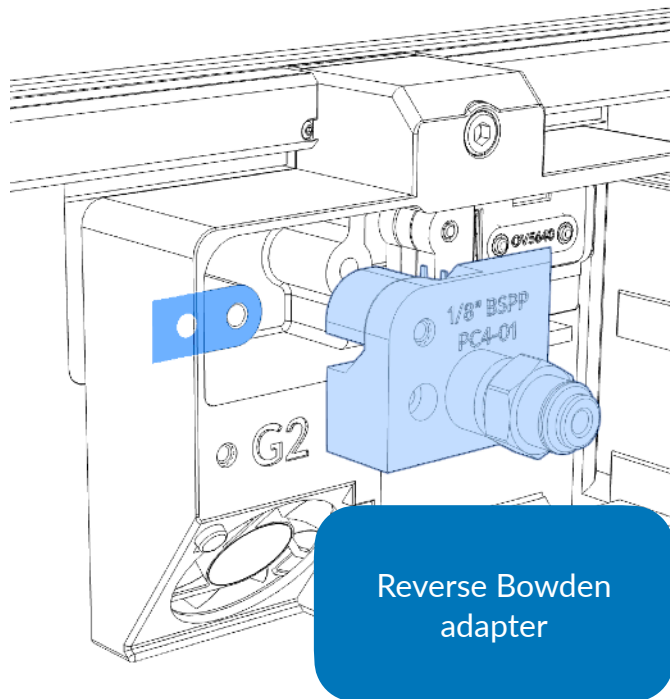
Pay attention to the
wires between the
LED units.



M5*40mm SHCS



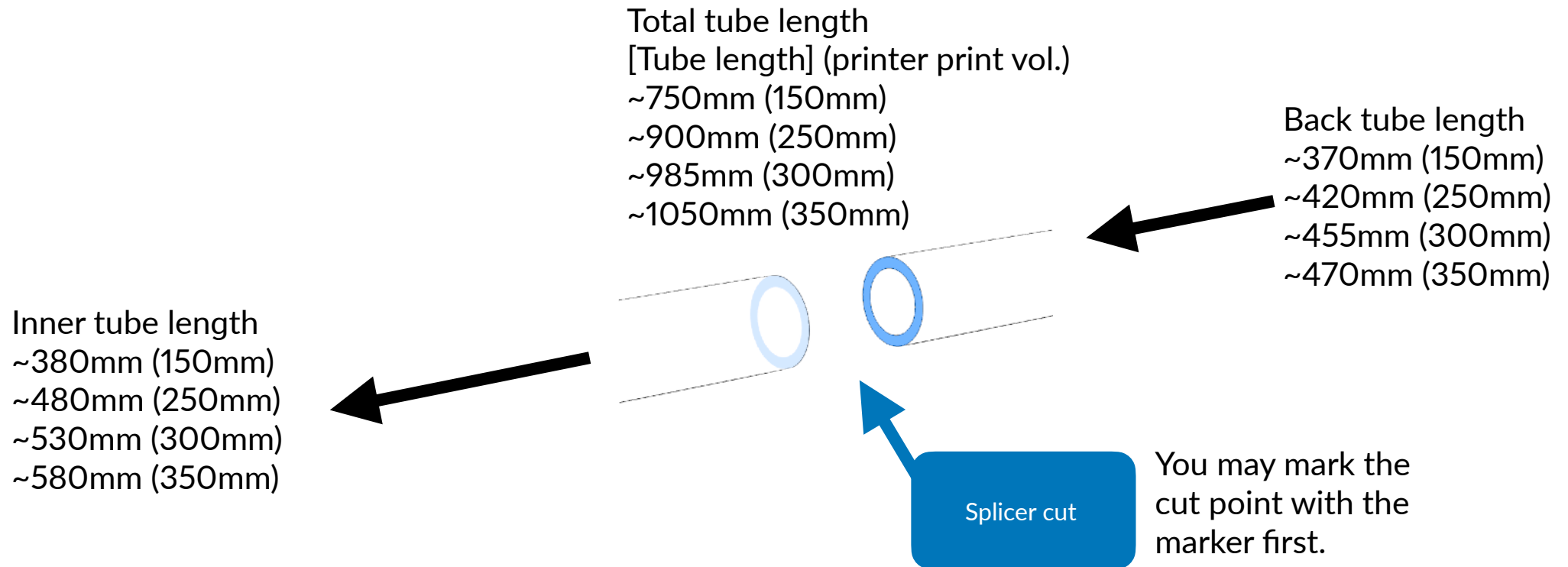
Attaching the reverse Bowden adapter



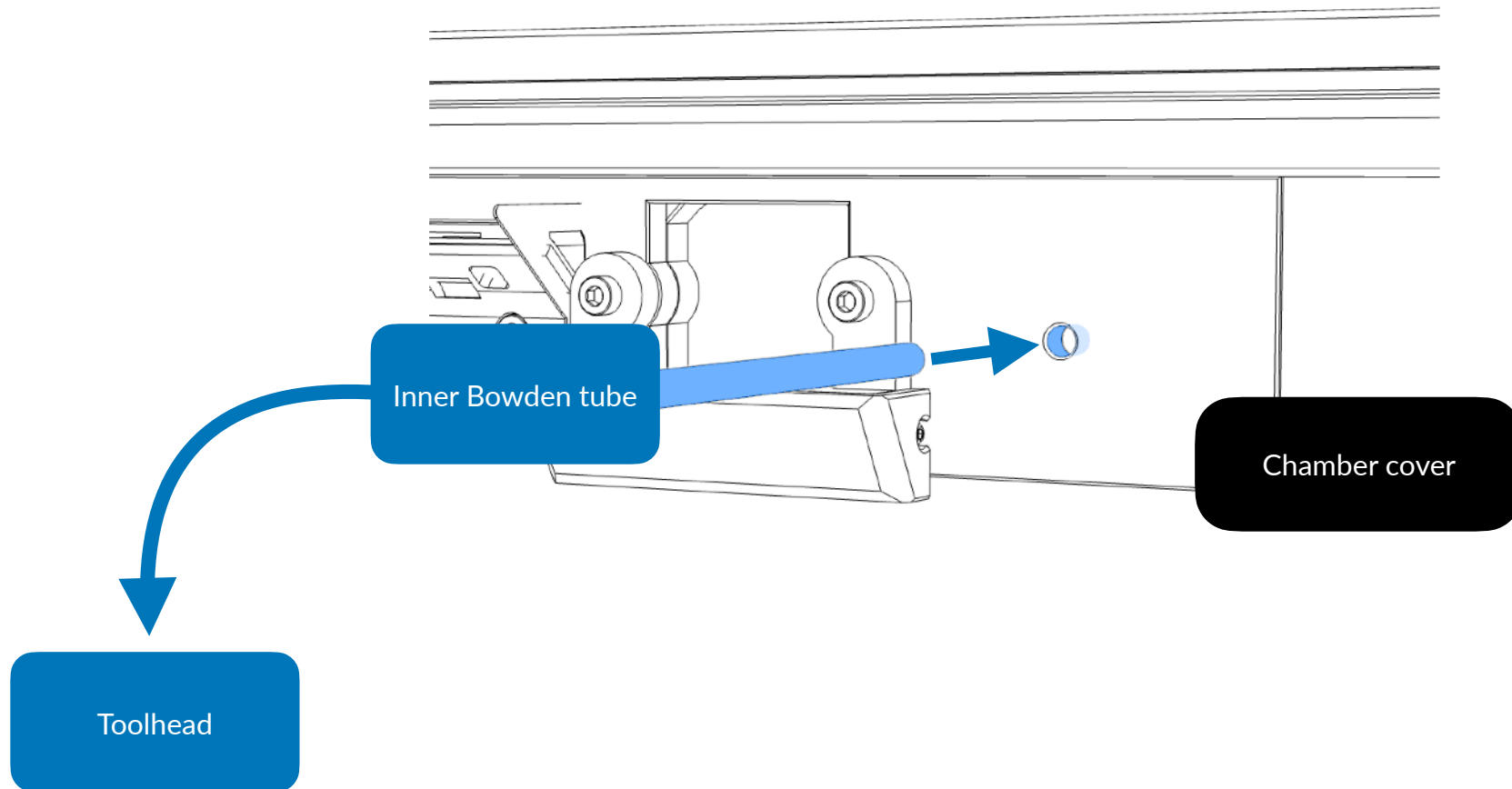
Attaching inner Bowden tube

Preparation

You may need a spare tube if your tubes are too short. You can refer to the total tube length if you are planning to replace the Bowden tube.



Attaching the inner tube

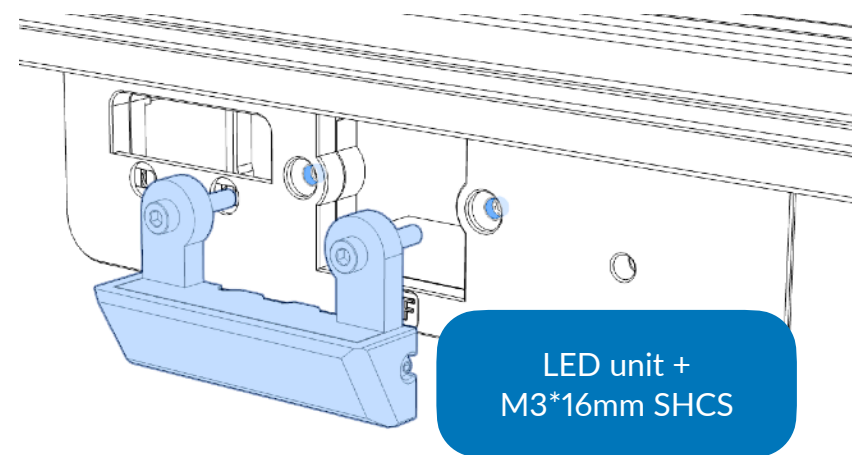
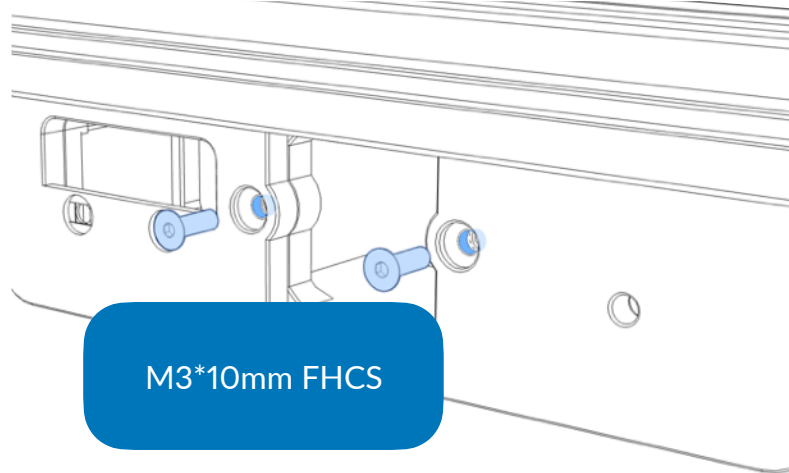


Inner screws mounting

Without LED unit

OR

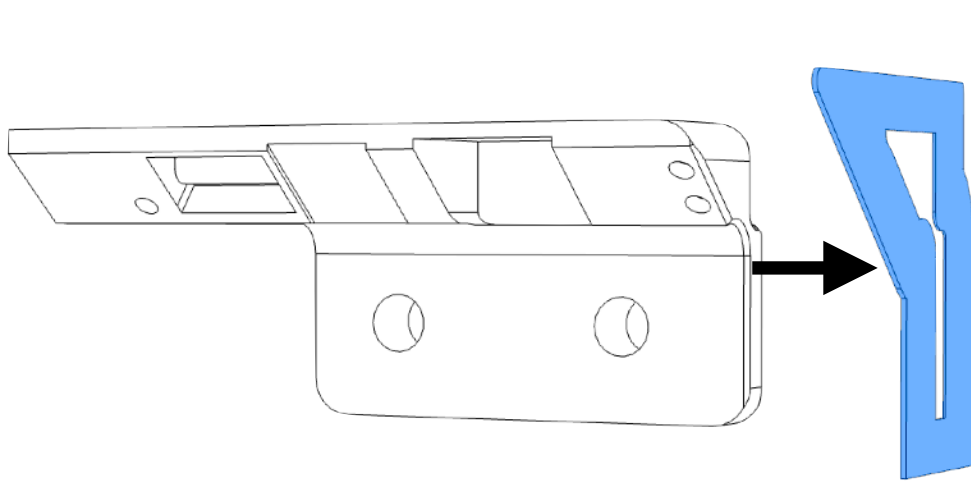
With LED unit



Webcam installation

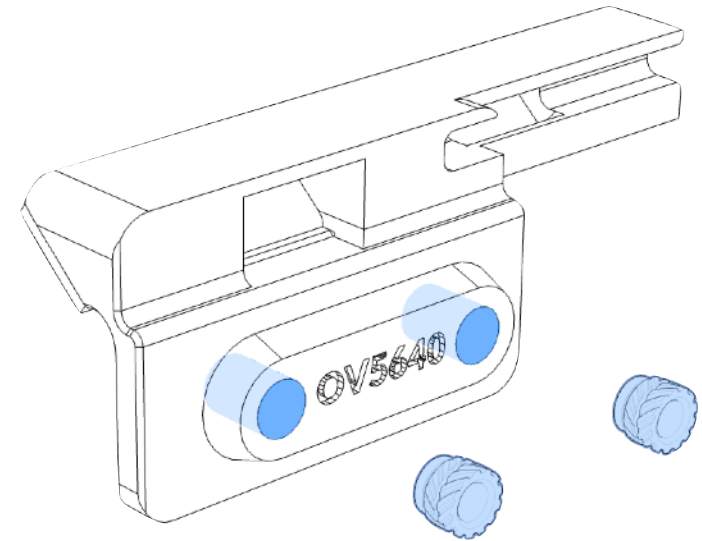
OV5640 laptop webcam

Preparation



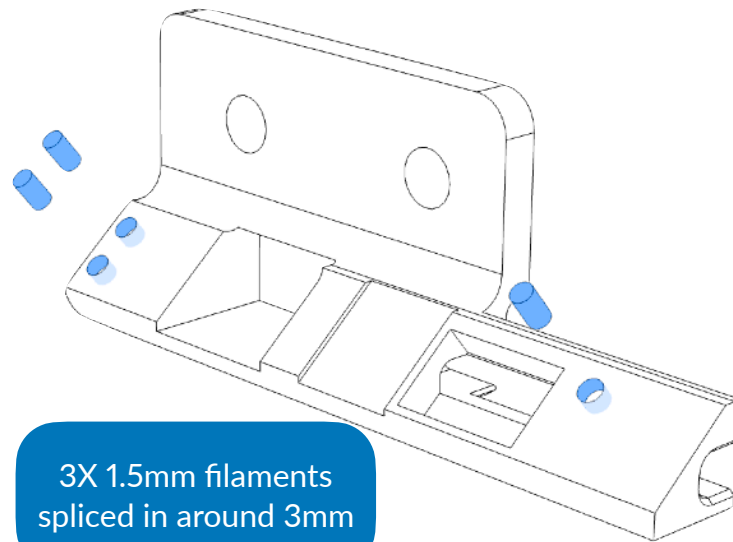
OV5640 camera mount
ov5640-mount-back.stl

In-model support base

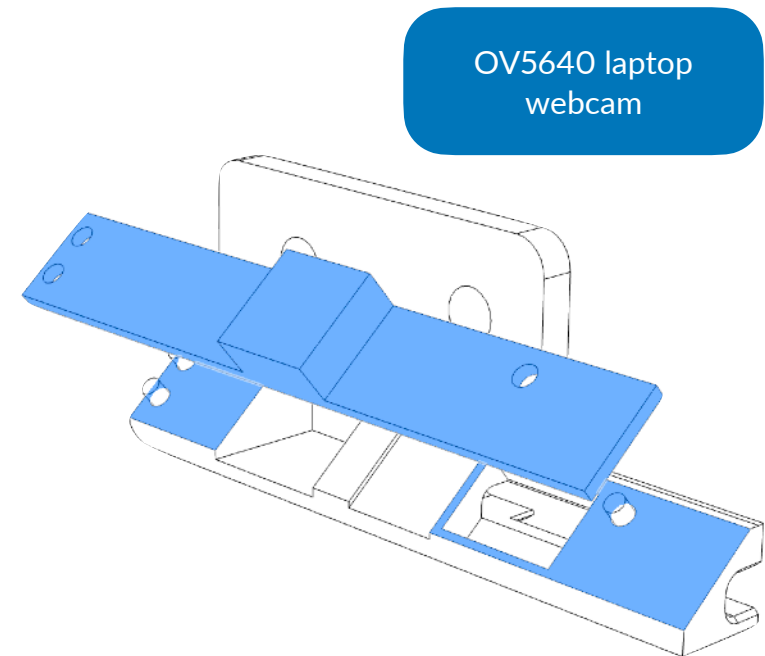


2X M3*5*4 heat set
inserts

Attaching the OV5640 laptop webcam

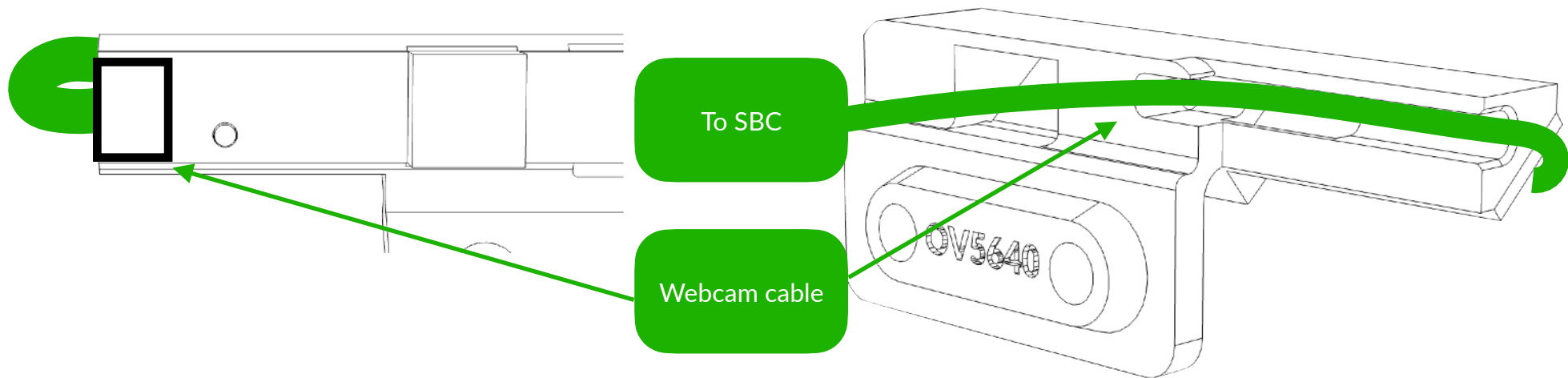


Note: if the filaments are too small, you may need to remove them while mounting to the exhaust unit. If the filaments are too big they cannot insert, this step should be skipped.



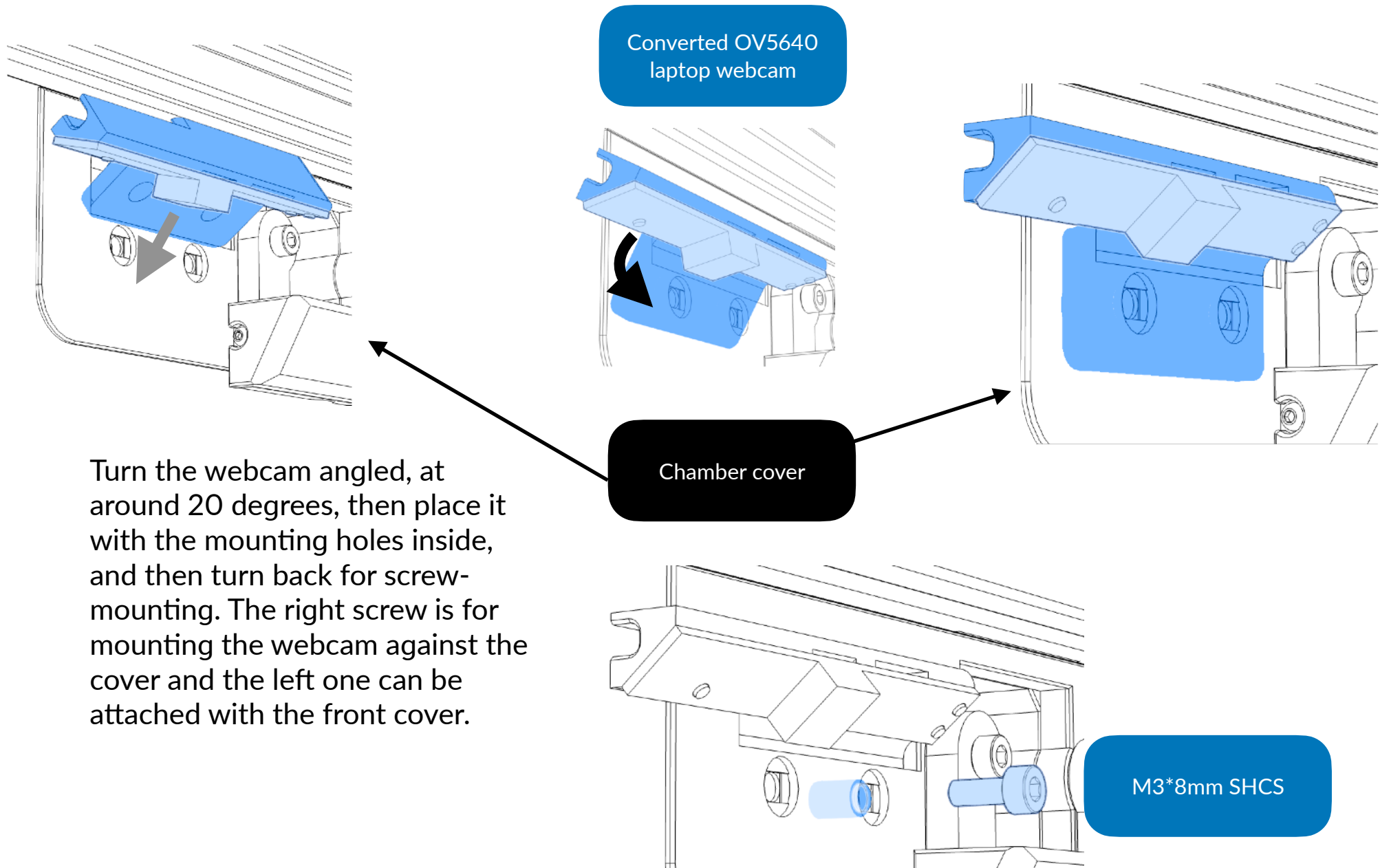
Some OV5640 webcams may come with back adhesive which you can use it to secure the webcam to the mount while the unit is being assembled.

Wire feeding and guiding to OV5640 laptop webcam mount



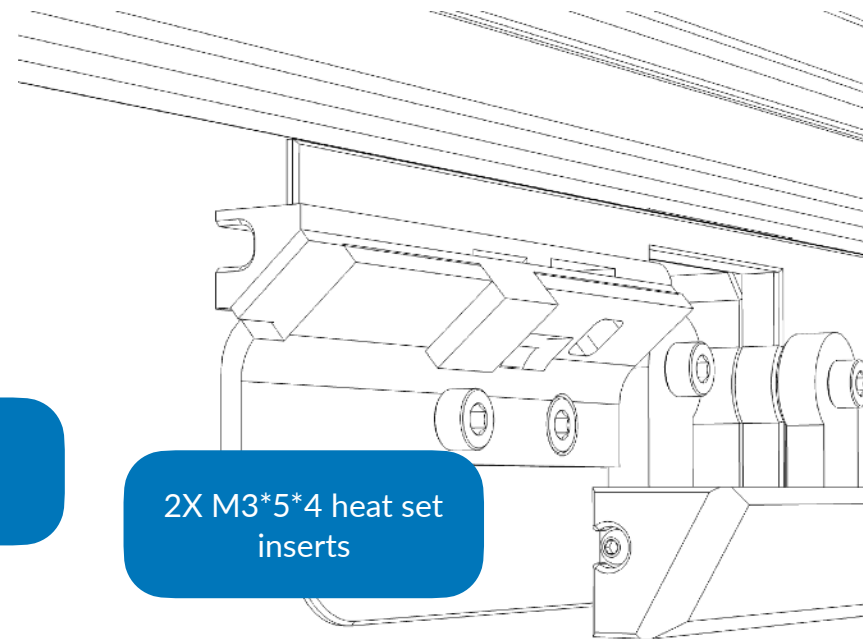
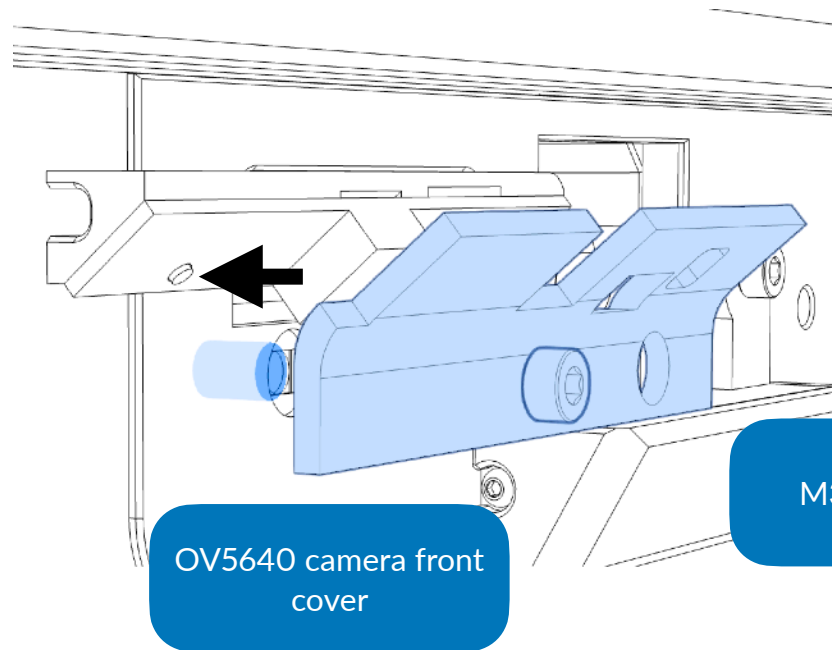
The model comes with the cutout that is just for wire guiding. To use it, you must connect the webcam cable first, and then guide the wire before and after installation.

Mounting converted OV5640 laptop webcam to ELG2

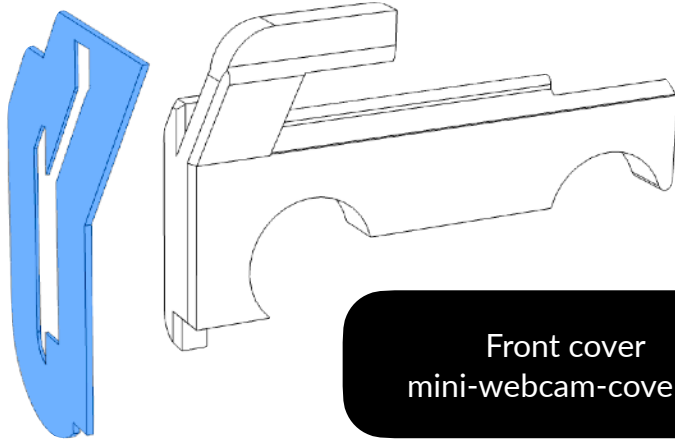


Attaching OV5640 laptop webcam cover

Partially assembled mini
webcam with screw mount



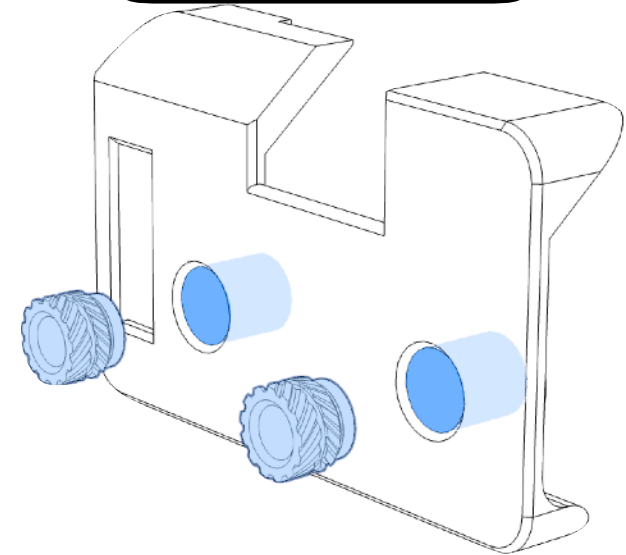
Mini webcam Preparation



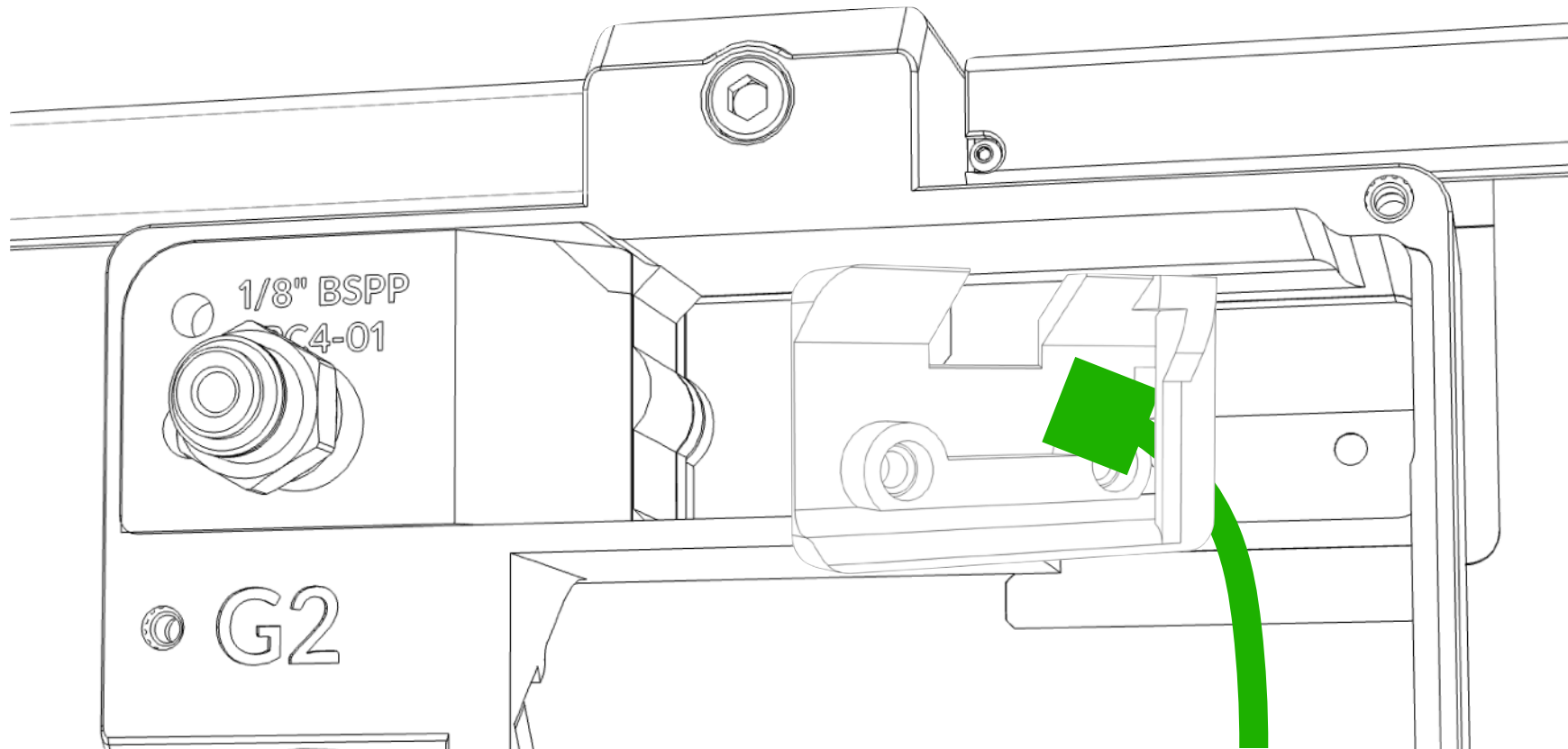
Front cover
mini-webcam-cover.stl

In-model support base

Mini webcam mount
mini-webcam-backside.stl

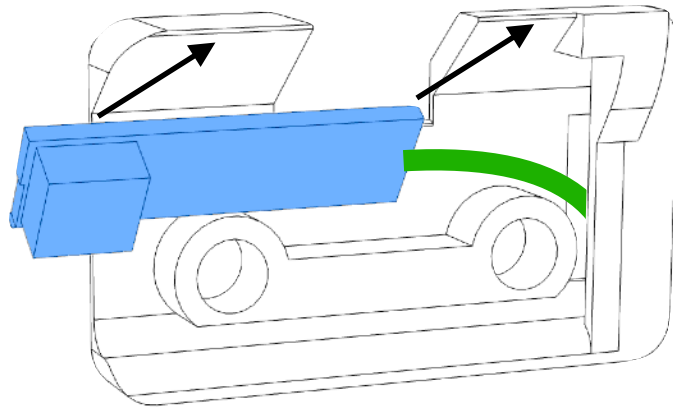


Feeding cable to mini webcam mount



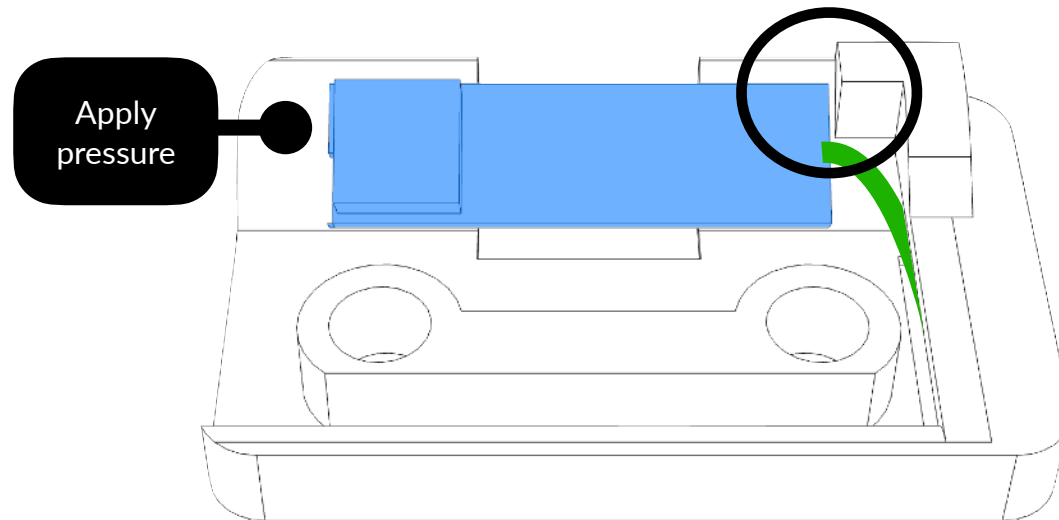
Un-feed the cable from the webcam cutout of ELG2 cover, then pass through the cable to the mount while the cable's webcam connector is still in the body unit.

Attaching the mini webcam to the mount



Connect the webcam cable to the mini webcam, then use the webcam's adhesive.

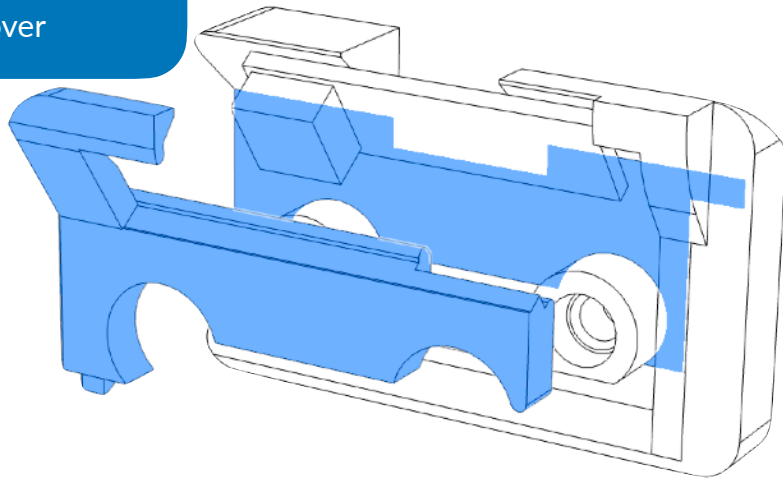
Mini webcam



When attaching the webcam, it needs to be aligned righthand. Otherwise, the front cover will not place properly.

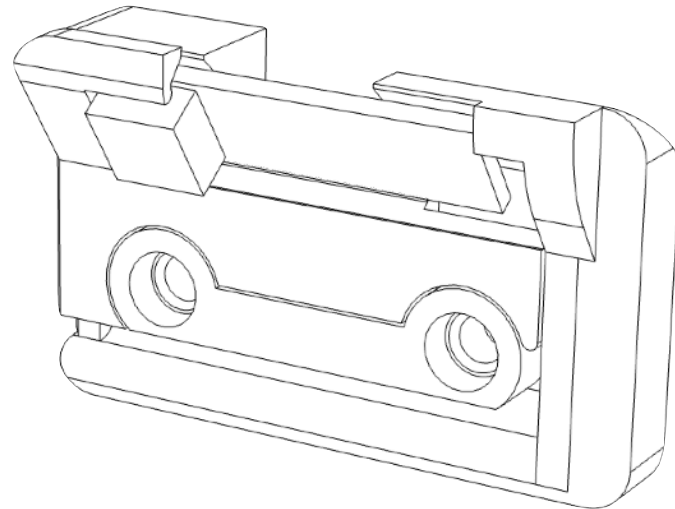
Attaching the mini webcam cover

Mini webcam front
cover

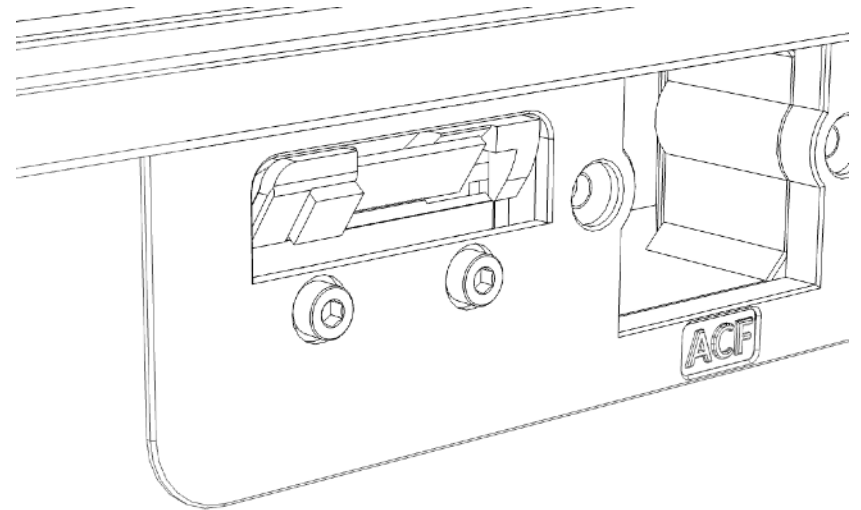
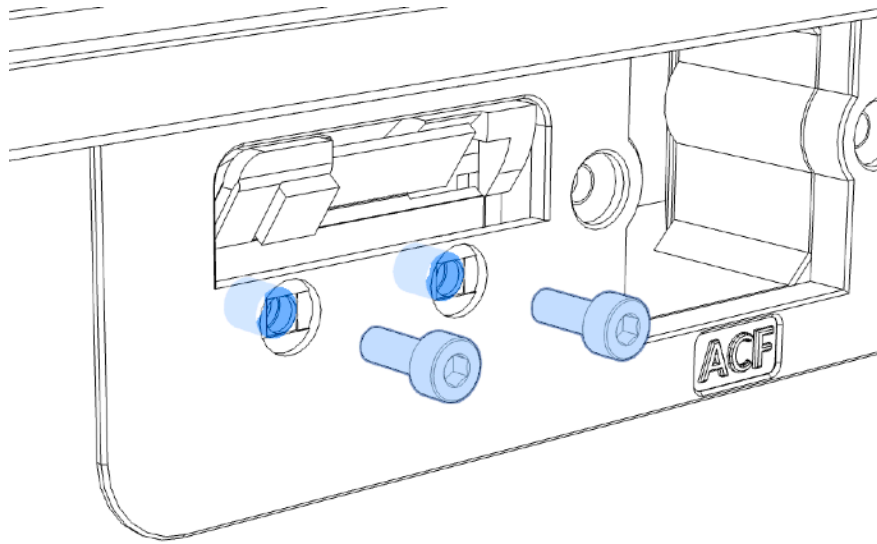


2X M3*8mm SHCS

Chamber cover



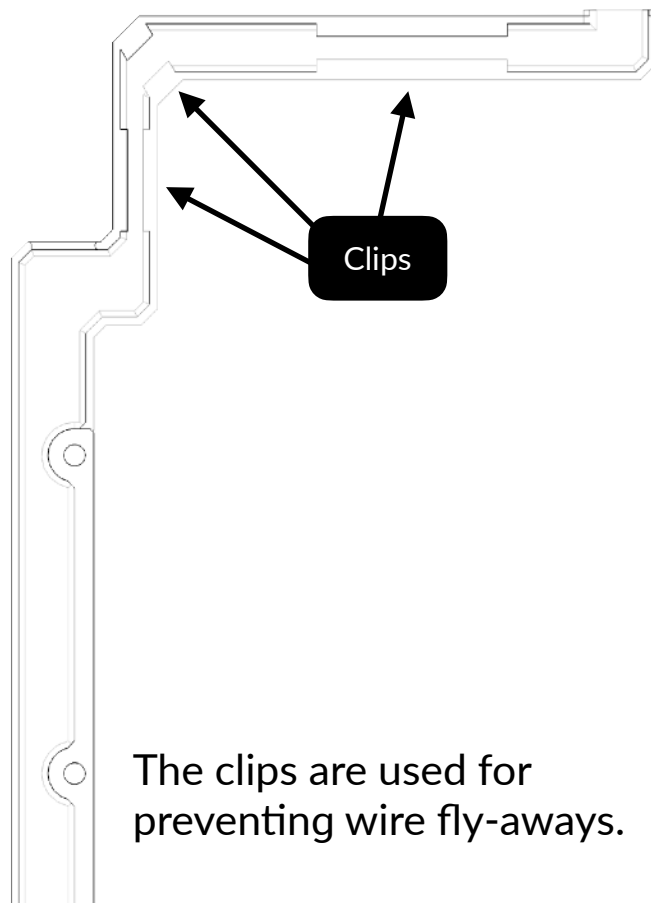
Attaching the mounted mini webcam to chamber cover



Cable cover

There are 4 models for 2 variants of panel thickness (4mm and 6mm) and the maximum space thickness (10mm and 15mm). You can choose the thinner version (10mm) for better aesthetics or thicker version (15mm) for extended cabling space. You may remove the normal middle panel clip of the right side of the back panel.

Tucking wires

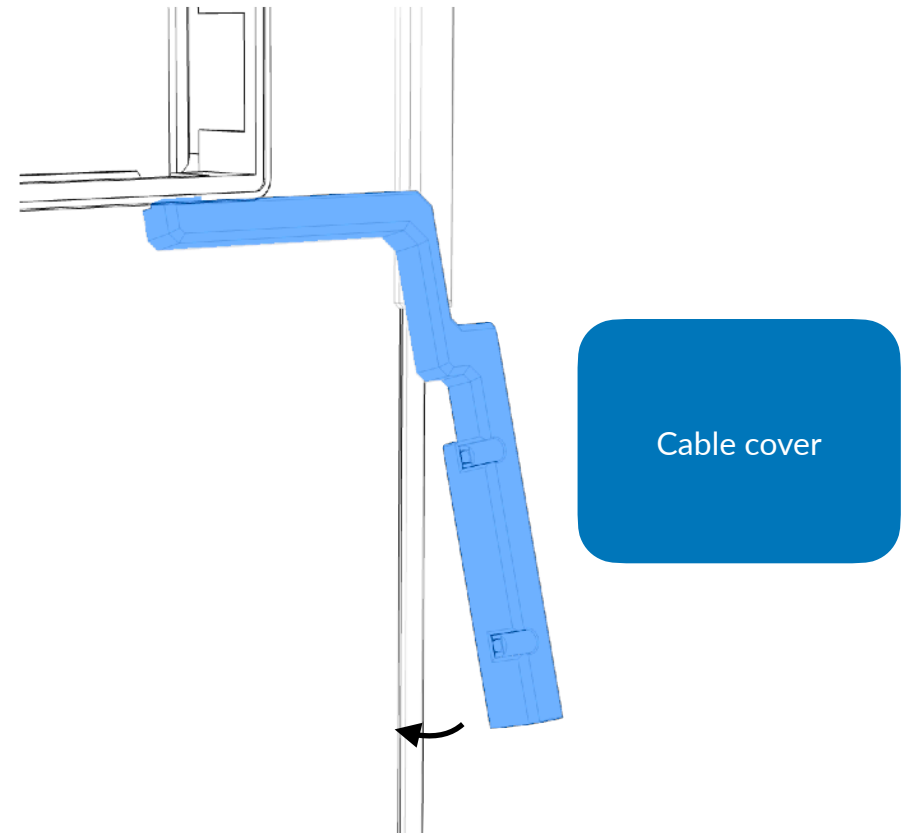
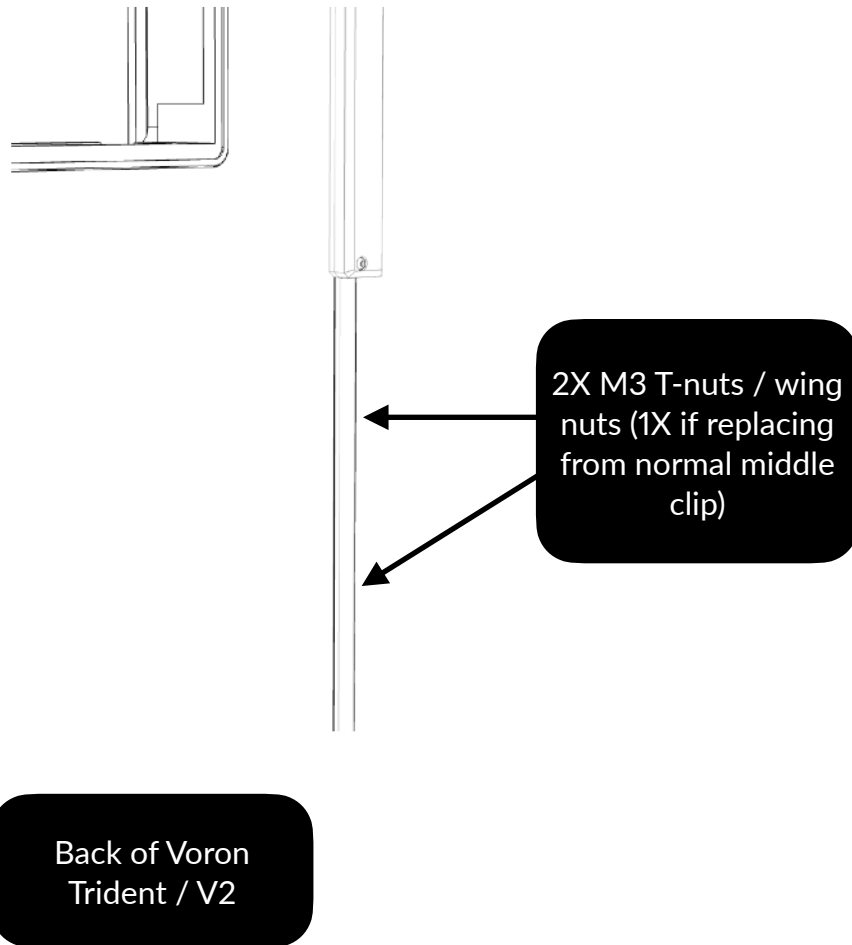


Variants:

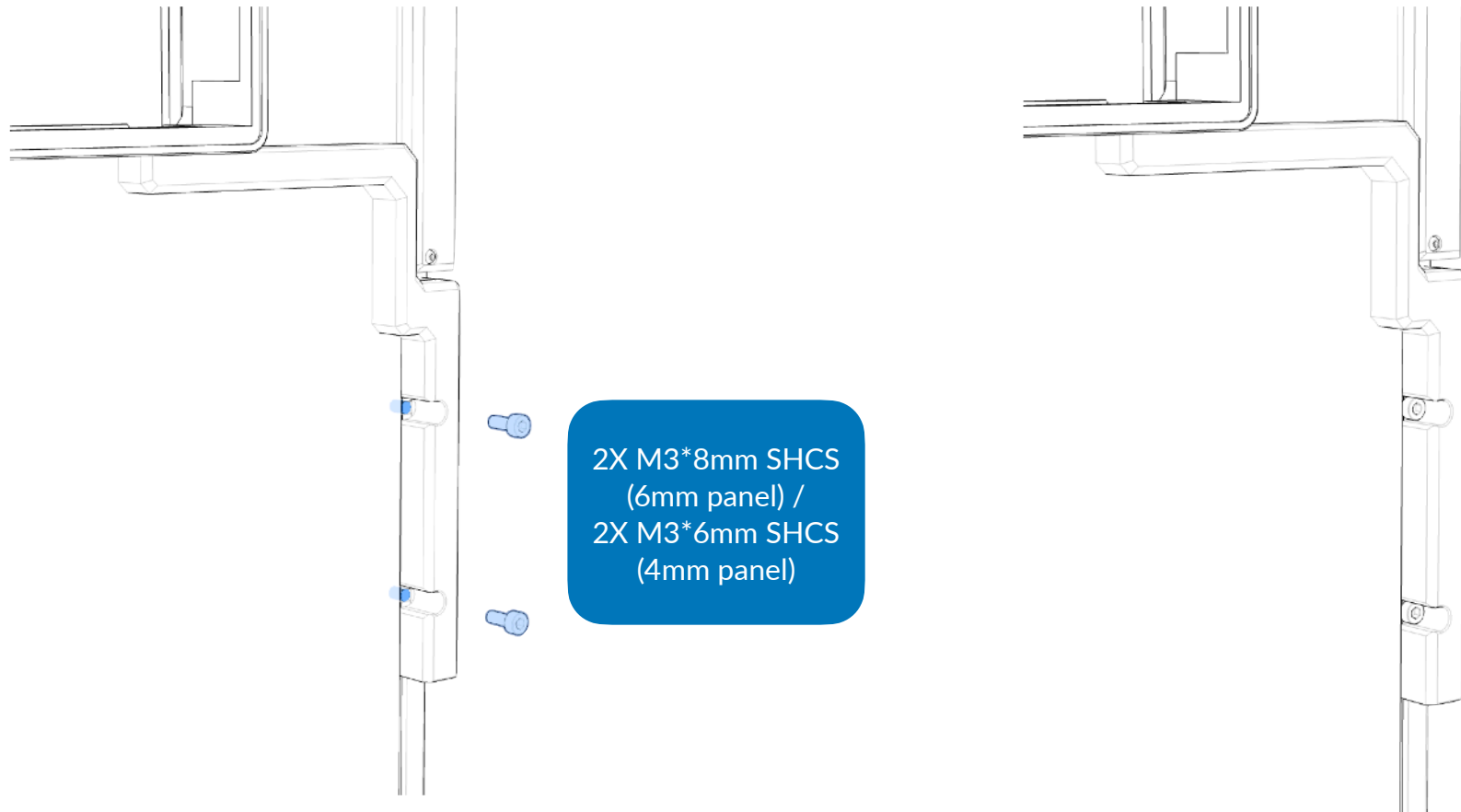
- cable-cover-unified-15MM-4mm (4mm panel, 15mm total thickness)
- cable-cover-unified-10MM-4mm (4mm panel, 10mm total thickness)
- cable-cover-unified-15MM-6mm (6mm panel, 15mm total thickness)
- cable-cover-unified-10MM-6mm (6mm panel, 10mm total thickness)

Attaching the cable cover

T-nuts must be preloaded.



You may want to place it angled, and tuck wires back in. You can move it up or down when T-nuts need to be aligned.

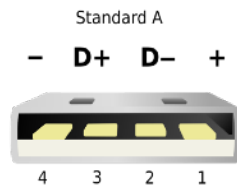


Once that's done, you should feed the wires to the controller units.

Webcam wiring

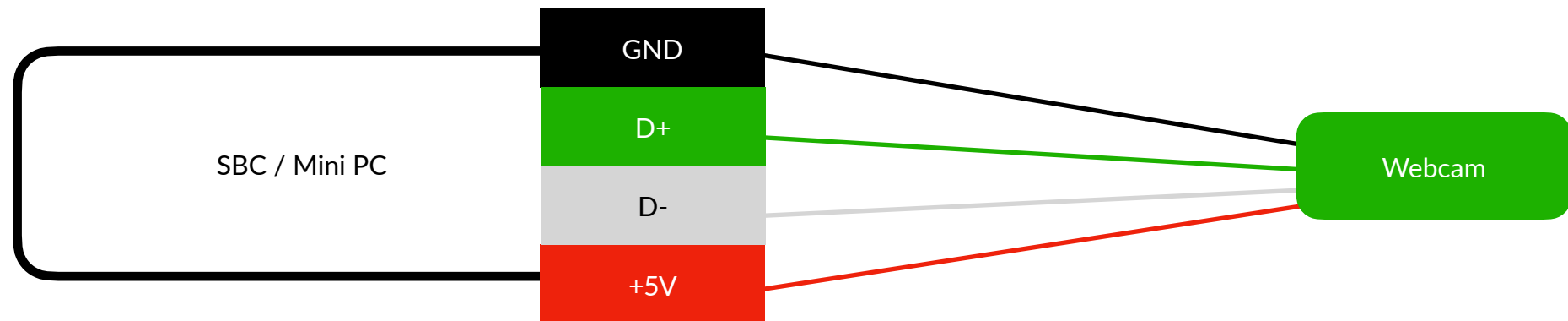
Wirings are illustrated

Unless if you are using a cable that's long enough, this might be a step that may take some time.

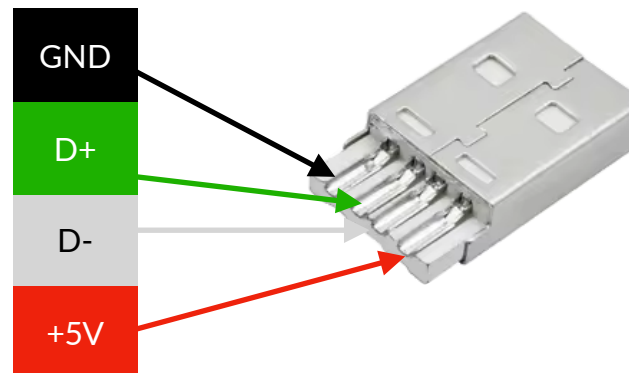


(Plug contacts facing up)

Simon Eugster - Simon / ?! 19:02, 7 January 2008 (UTC) - Own painting/graphic, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=3353998> <https://commons.wikimedia.org/wiki/File:USB.svg>

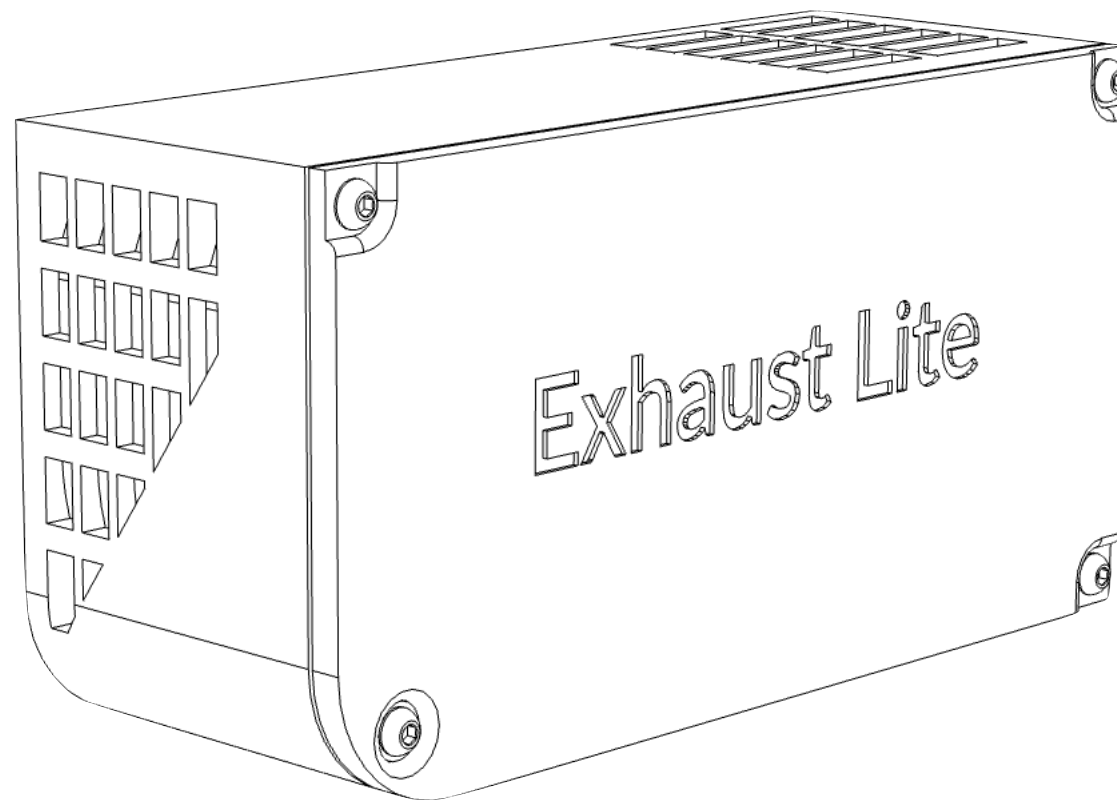


However, there are plugs that have it's contacts facing down but the solder points are inverted. This means that the contact points needs to be inverted as well, to prevent reverse polarity damage.



https://www.alibaba.com/product-detail/USB-Connector-A-Male-seat-4_62245238606.html?spm=a2700.7724857.0.0.3fdc24f8ml7VR0

Filter cartridge



Preparation

This is the most important part of Exhaust Lite. Without it you are exhausting with 100% dirty air to outside.

Filter specs

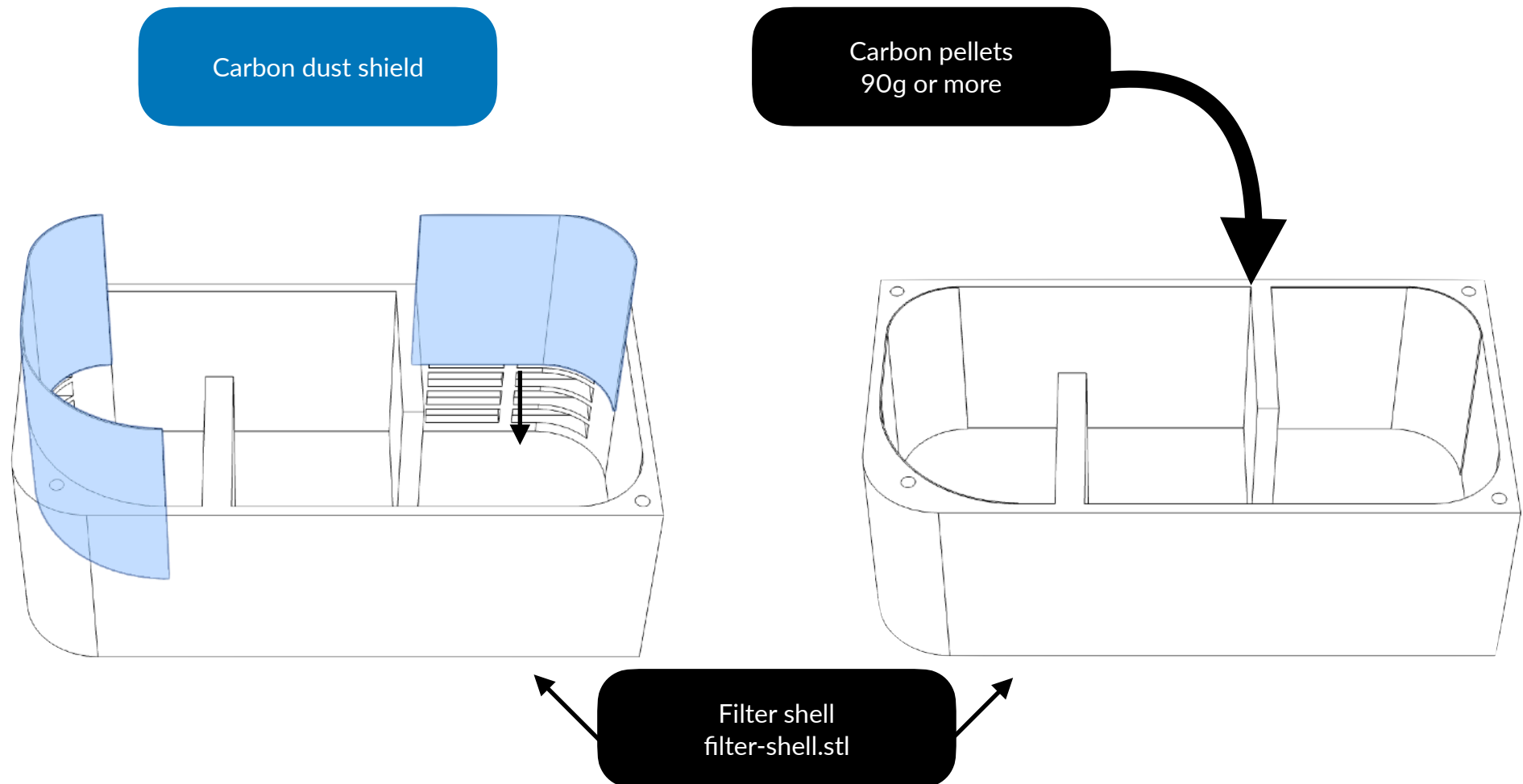
ELG2 filter needs to be filled with activated carbon pellets before being used. Otherwise, it will exhaust with 100% dirty air.

Each filter cartridge requires at least 90g of carbon pellets.

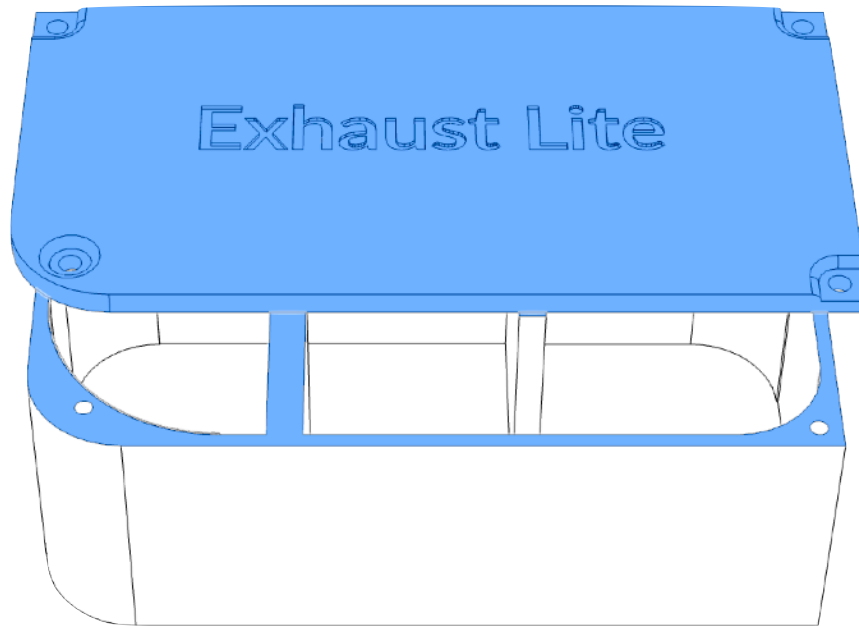
The filter against the carbon dust is recommended. You can make 30x40mm and 30x67mm shield for intake (top-right) and exhaust (left) respectively. You may use new filters. You may need to use the double-sided adhesive to stick the shield to the sides of the unit.

Spec	Minimum	Recommended
Carbon type	4mm Pellets	
Carbon acidity	Neutral (pH 7-9)	
Carbon Tetrachloride	60% or more	80% or more
Effective area	600m ² /g or over	1000m ² /g or over

Filter cartridge assembly

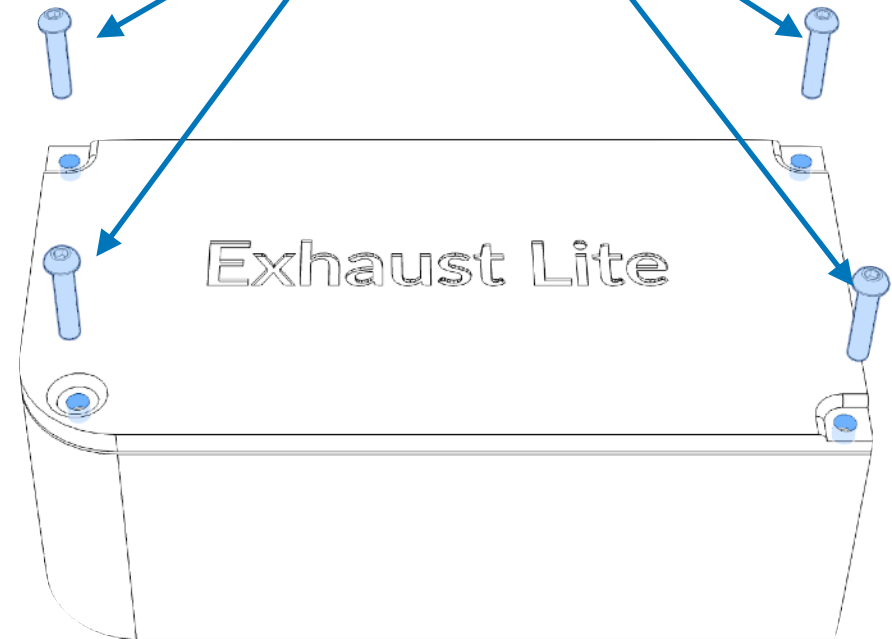


Filter cartridge top cover
filter-cover.stl



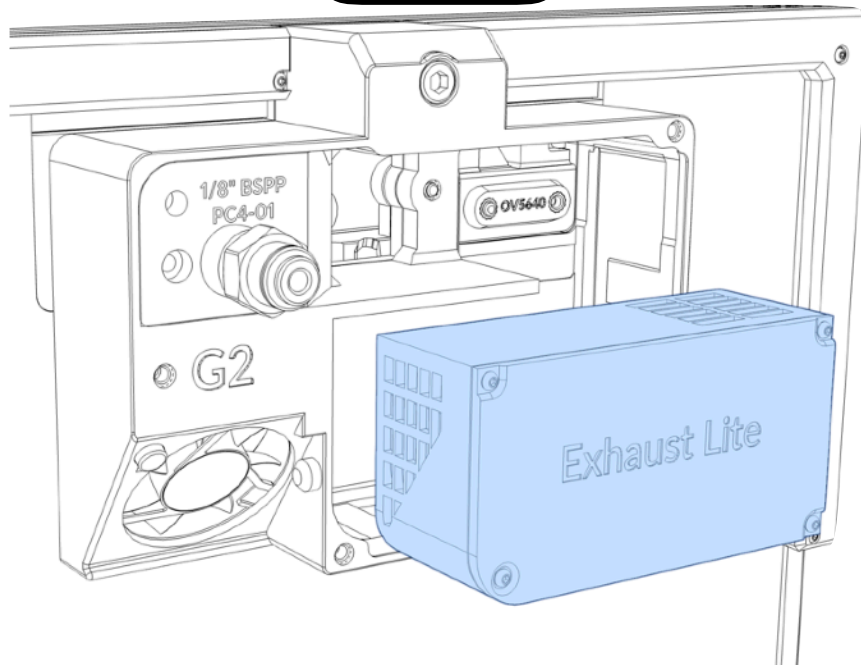
Completed filter shell

4X M2*10mm self-
tapping screws

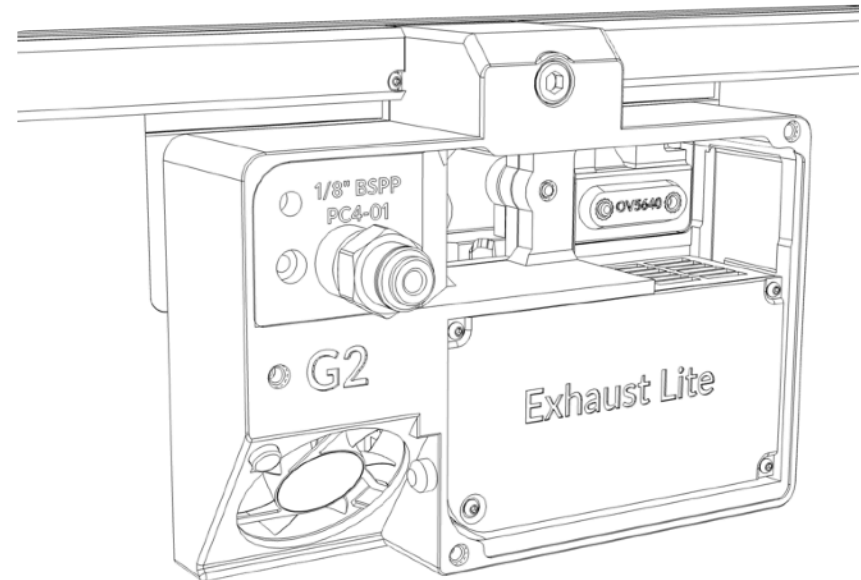


Slotting the filter to the body

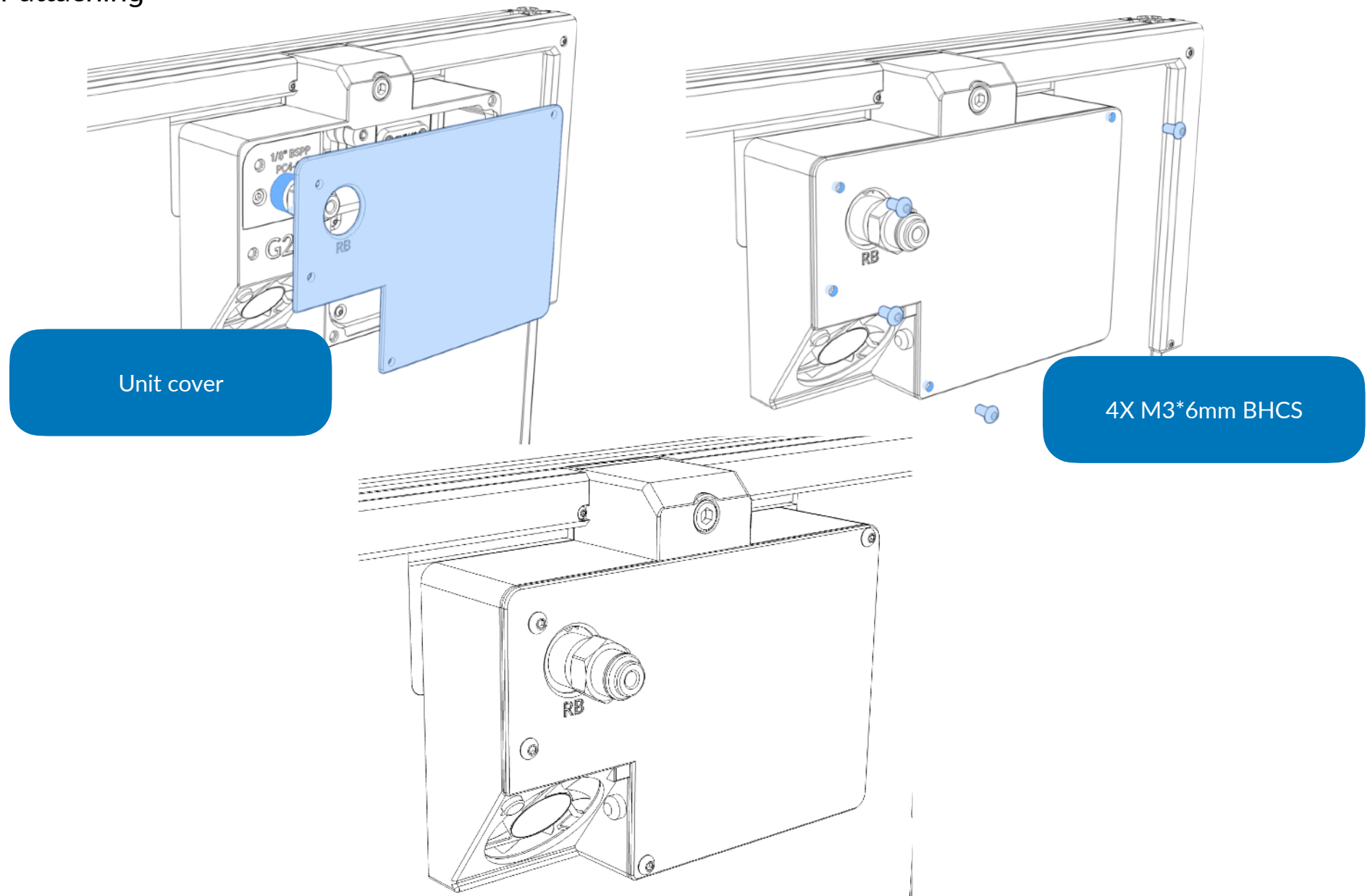
Body unit



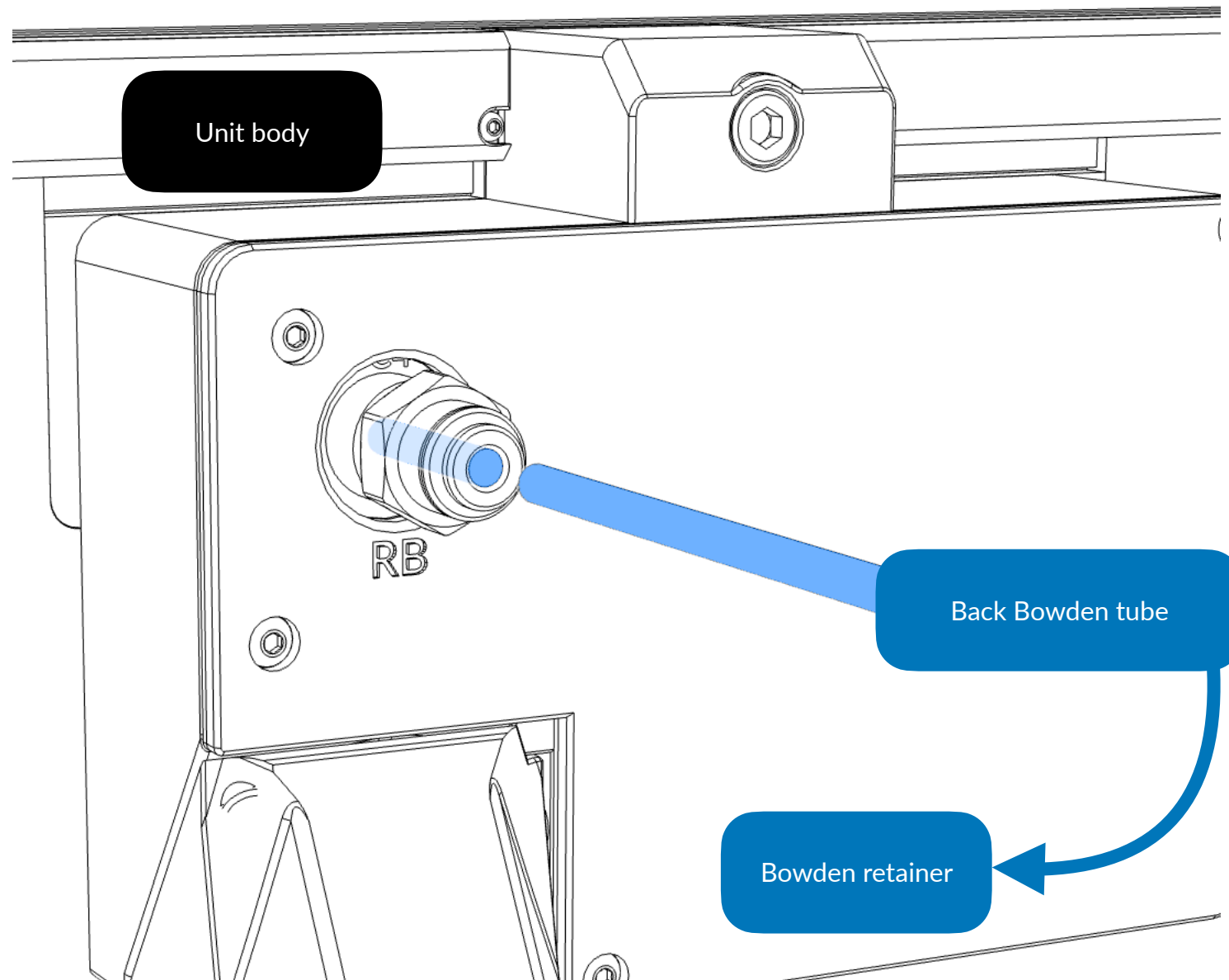
Assembled filter cartridge



Cover attaching



Attaching the back Bowden tube



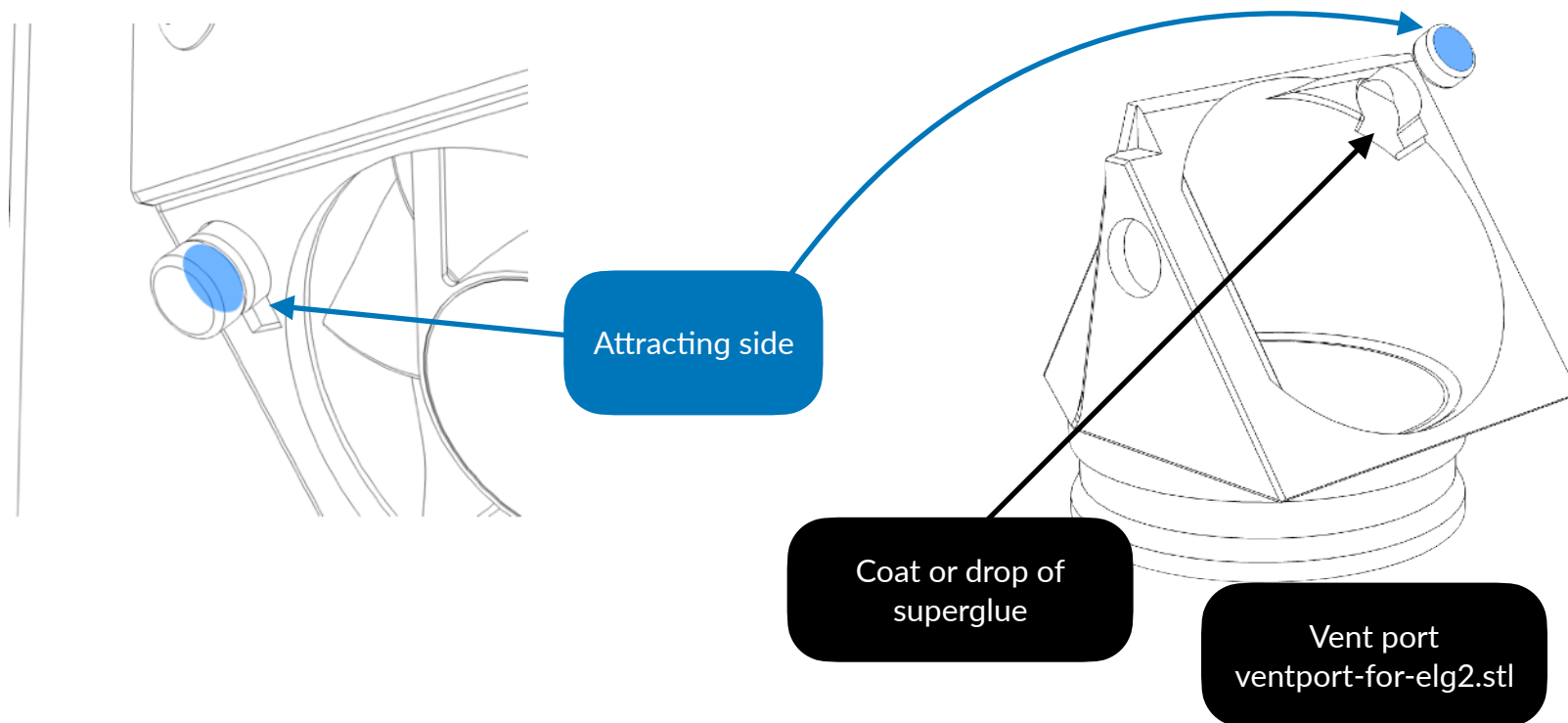
Vent port

Preparation

The 6*3mm magnets must attract each other for the vent port to be attached.

If they repel each other, the vent port will never be attached.

The vent port's magnet slot must be coated with or with at least a drop of superglue.



Software configuration

Some settings for the original exhaust unit can be kept unchanged. However, for the LEDs, you may have to change, or insert if it is connected to the different pin. You may refer to Config/Exhaust Lite for the code and they will be explained. For the runout sensing, you may refer to the dedicated runout.cfg configuration file.

To use ELG2 configuration files conveniently, you can download elg2.cfg and runout.cfg files, then put them into your printer's configuration window and then add [include elg2.cfg] into the base configuration file (mostly printer.cfg) and it should ready to go.

If your printer has never used exhaust fan before, you may try finding exhaust fan config in your base config file and uncomment first. Otherwise, if it is too impossible, you may uncomment that inside elg2.cfg configuration file.

runout.cfg is just a plug-and-play configuration file. If you have installed the coupler adapter that has filament runout sensing, you can uncomment the runout.cfg assignment [include runout.cfg] in elg2.cfg configuration file.

You should change the pin assigned if the board uses pin that is different, like for example BTT Manta M8P, Kraken and Makerbase Monster8.

Assembly complete

And we are done! If you are satisfied with such product consider tagging @officialbunny350 on Facebook or @realBunny350 on X (formerly known as Twitter) and share such product with us on Oitswilliam Pang Discord Server.

If you have enquires, problems or issues related to building or using such product, please report it on [GitHub.com/Bunny350/OITSWILLIAMV2/issues](https://github.com/Bunny350/OITSWILLIAMV2/issues) and we will get on it.

See you guys next time!

Change log

12042024 - Changes to 1/8" BSPP with filament runout detector. Added the second ball bearing in an attempt to address part grinding, especially when used abrasive filament on such weaker part. Add recommended wrench size for coupler.

05202024 - Manual release for ELG2, after many months of wait and finale refresh.

It's done. While you are at my social
platforms maybe should you follow me?

Manual designed by Oitswilliam Pang.
One-person operated design.
www.oitswilliam.com